



SPSS

(Statistical Packages for the Social Sciences)

SHORT INSTRUCTIONS

This presentation contains only relatively short instructions on how to perform basic statistical calculations in SPSS. Details around a certain function/analysis method not covered by these instructions are often more or less intuitive and self-explanatory. There is also a Help button in every dialog window that you can use to get more information.

TUTORIAL

There is a step-by-step tutorial available in SPSS, you can find it by clicking **Help >> Tutorial** in the Menu bar. It will also show as one of the possible options at startup.

BE CAREFUL

Statistical software has very limited possibilities to critically review the information that is being entered, and the results being processed. It is therefore of utter importance to keep track of which assumptions need to be fulfilled in every situation, and how the results should be interpreted.

Please send an e-mail to inger.persson@statistik.uu.se if you discover anything that is incorrect in this document.

Updated 2014-10-08

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1 Online introductions and manuals

IBM SPSS Statistics 22 Brief Guide (98 pages) describes how to; open and import data files, enter data, edit data, produce summary statistics and some graphs, etc.

ftp://ftp.software.ibm.com/software/analytics/spss/documentation/statistics/22.0/en/client/Manuals/IBM_SPSS_Statistics_Brief_Guide.pdf

IBM SPSS Statistics 22 Core System User's Guide (286 pages) describes how to; open, import, and export data files, edit and transform data, create pivot tables, etc.

ftp://ftp.software.ibm.com/software/analytics/spss/documentation/statistics/22.0/en/client/Manuals/IBM_SPSS_Statistics_Core_System_User_Guide.pdf

IBM SPSS Statistics Base 22 (198 pages) describes how to; produce descriptive statistics,

crosstabs, explore data (including Normality plots), perform t-tests, calculate correlations, linear regression, and much more.

ftp://ftp.software.ibm.com/software/analytics/spss/documentation/statistics/22.0/en/client/Manuals/IBM_SPSS_Statistics_Base.pdf

You can also find introductions to SPSS online, eg. this one (at YouTube):
<http://www.youtube.com/watch?v=eTHvIEzS7qQ> (approx. 10 minutes)

2 Installing SPSS on your own computer

If you wish to install SPSS on your own computer you can download a free 14-day Trial version here:

http://www14.software.ibm.com/download/data/web/en_US/trialprograms/W110742E06714B29.html

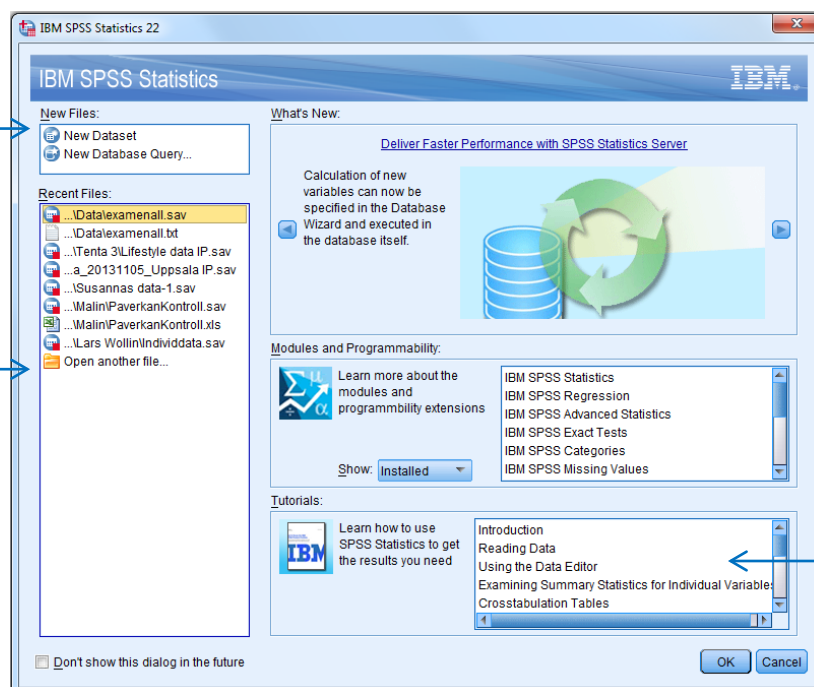
There are also student licenses available, 6 or 12 months. “SPSS Statistics Base GradPack” is sufficient for the first course in Quantitative methods. Logistic regression is however not included, for that you need to choose “SPSS Statistics Standard GradPack”.

3 Opening SPSS

When opening SPSS from the Start menu the following window should appear.

Choose “New Dataset” if you want to open an empty data set, either to manually enter data or to import data from eg. Excel (see sections 6, 7, and 8)

Find your data source here if you have an existing SPSS data set



Run a tutorial if you want to learn more about SPSS

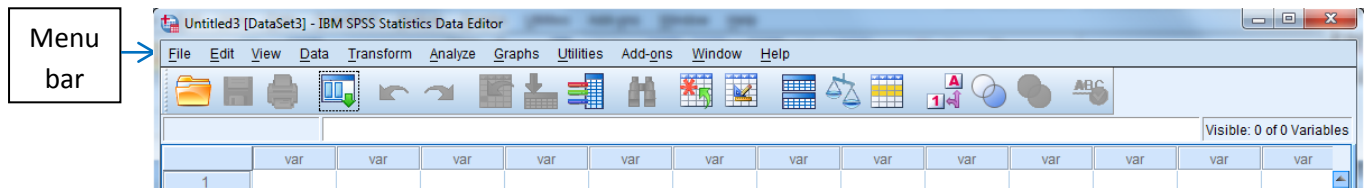
4 The different windows/views of SPSS

There are two different windows in SPSS, Data Editor window and Output window. When

you open SPSS, the Data Editor Window will appear (see section 4.2).

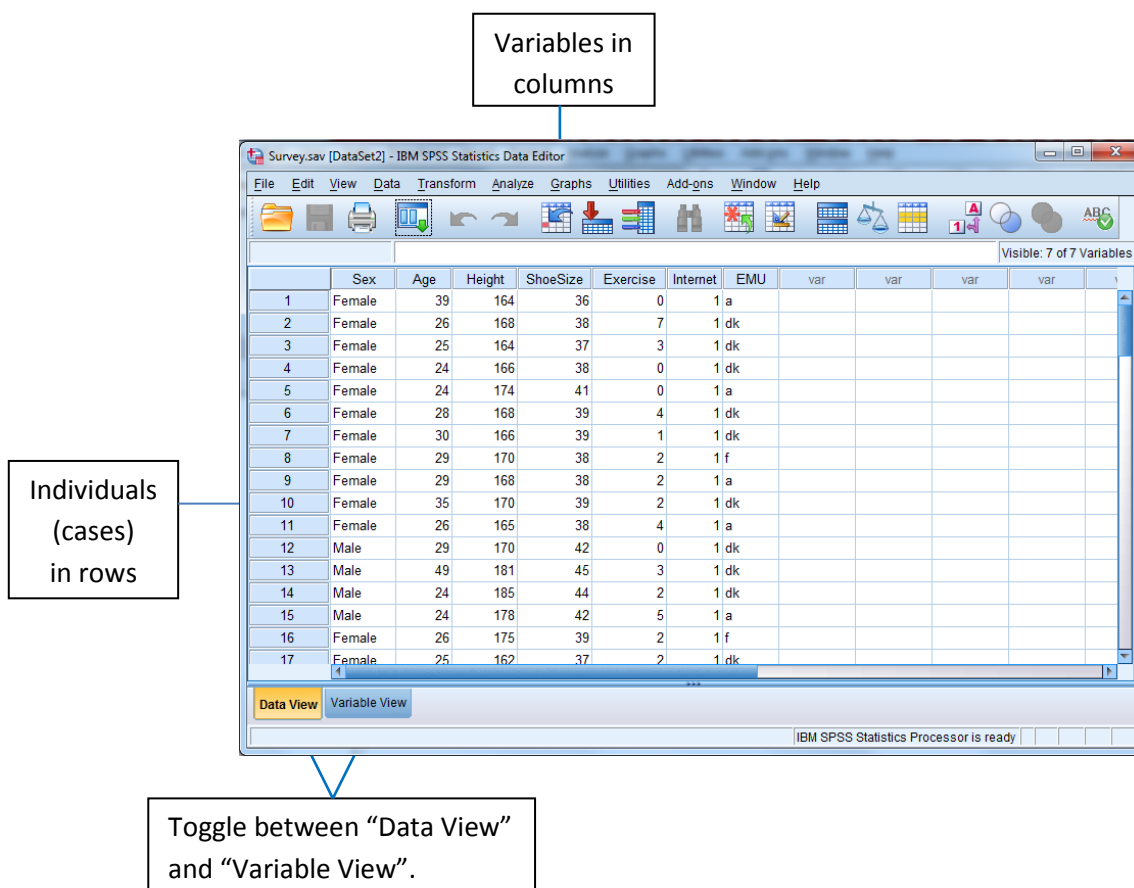
The Data Editor window has two different views, Data View and Variable View, described in sections 4.1 and 4.2 below. The Output window is described in section 4.3 below.

The options of the Menu bar in the Data Editor Window are also included in the Output window, so you can perform all statistical procedures from any of the windows.



4.1 Data View (Data Editor window)

In Data View the variables are displayed, with their names and variable values for each individual (case).



In SPSS (as in all statistical software) individuals/cases are represented by rows in the data set, and variables by columns.

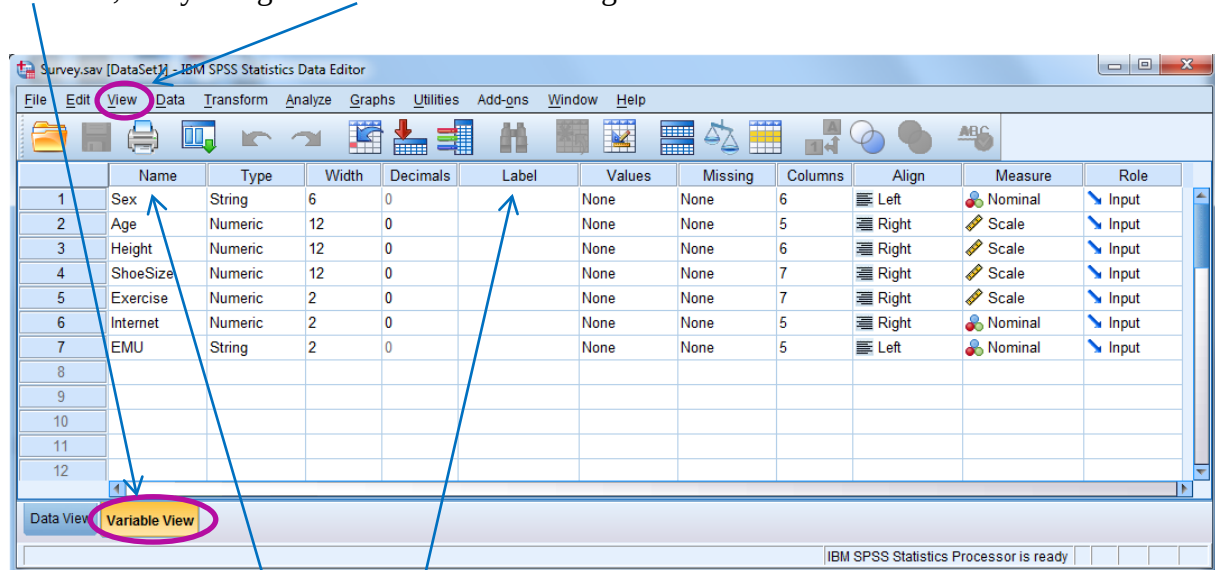
Example above: Row 10 contains data for a female, who is 35 years of age, 170 cm tall, with shoe size 39, and so on.

Most statistical analyses are performed on variables, i.e. columns.

4.2 Variable View (Data Editor window)

In Variable View the properties of each variable are displayed.

You can find the Variable View either by clicking on “Variable View” at the bottom of the window, or by using the Menu bar and clicking **View >> Variable**.



4.2.1 Name variables

The variable name is the name used by SPSS to identify the variable. To name a variable, click the box under “Name” and type the desired name for each variable. The name can be up to 64 characters long. Variable names cannot contain blank spaces, and should start with a letter. Letters, numbers, underscore (_), period (.) etc are allowed.

4.2.2 Define variable labels

A variable label is the text that will be displayed in any analysis output. Variable labels can contain a larger number of characters than the variable names, and also blank spaces, etc.

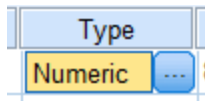
Click the box under “Label” and type the desired label for each variable.

NOTE! Variable labels are very useful. If you define them once you will get the correct description of your variables in all analysis output (e.g. including units!).

4.2.3 Define variable types (numeric, string, etc.)

SPSS uses the variable type to select which variables that can be used for which statistical

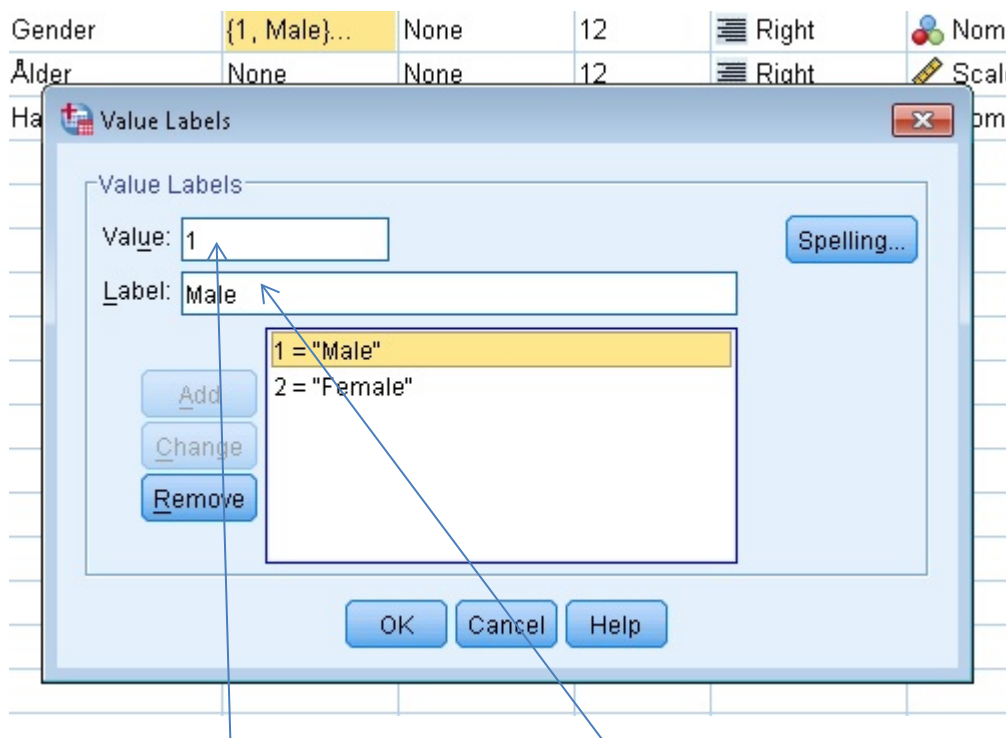
analysis methods. To change the variable type, click the box under “Type” and then click the blue square that appears. Select the appropriate variable type, “Numeric” if your variable values are numbers or “String” if the values are letters, and click “OK”.



4.2.4 Define value labels (using the label “male” for the value 1, etc.)

A value label is the label for a coded variable in the dataset. For example, “Gender ” may be coded 1 = Male and 2 = Females.

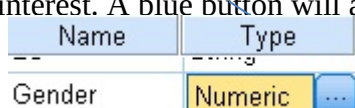
To add a value to your variable, click the box under “Values” that corresponds to your needed variable. The following window will then appear.



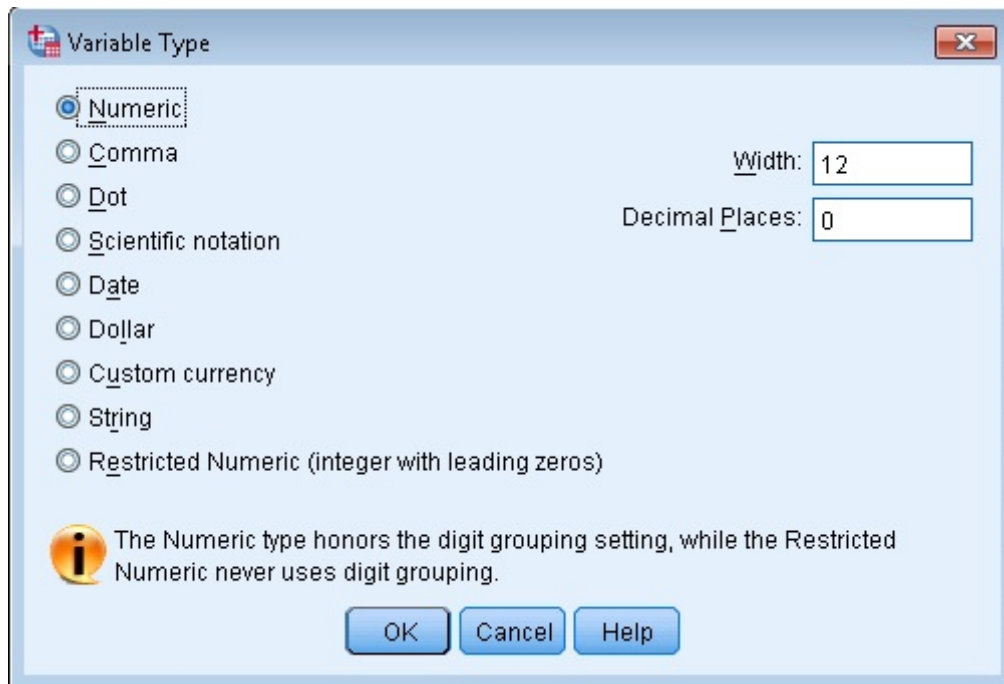
In the “Value” box add the value, in the “Label” box add the corresponding label to your value. The values can also be changed or removed in the same manner.

4.2.5 Define type of data (numeric, string etc.)

When defining the type of variable, you have to correctly identify the type of variable. SPSS has special restrictions in place so that statistical analyses cannot be performed on inappropriate types of data. Information for the type of each variable is displayed in the Variable View tab. Under the “Type” column, click the cell associated with the variable of interest. A blue button will appear.



Click the blue button and the Variable Type window below will appear. You can use this dialog box to define the type for the selected variable, and any associated information (e.g. width, decimal places).



The most used types of variables are numeric and string.

Numeric variables have values that are numbers (in standard format or scientific notation). Missing numeric variables appear as a period (i.e. “.”).

String variables, which are also called alphanumeric variables or character variables, have values that are treated as text. This means that the values of string variables may include numbers, letters or symbols. Missing string values appear blank.

Comma – numeric variables that include commas that delimit every three places (to the left of the decimals) and use a period to delimit decimals. SPSS will recognize these values as numeric – with or without period, and also in scientific notation.

Example: Thirty-thousand and one half: 30.000,50

Scientific notation – numeric variables whose values are displayed with an E and power of ten exponent. Exponents can be preceded by either an E or a D, with or without a sign, or only with a sign (no E or D). SPSS will recognize these values as numeric, with or without an exponent.

Example: 1.23E2, 1.23D2, 1.23E+2, 1.23+2

Date – numeric variables that are displayed in any standard calendar date or clock – time formats. Standard formats may include commas, blank spaces, hyphens, periods or slashes as space delimiters.

Example: Dates: 01/31/2013, 31.01.2013

Dollar – numeric variables that contain a dollar sign before numbers. Commas may be used to delimit every three places, and a period can be used to delimit decimals.

Example: Thirty-three thousand dollars and thirty-three cents: \$33,000.33

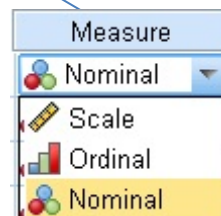
Custom currency – numeric variables that are displayed in a custom currency format, You must define the custom currency in the Variable Type window. Custom currency characters are displayed in the Data Editor but cannot be used during data entry

Restricted number – numeric variables whose values are restricted to non-negative integers (in standard format or scientific notation). The values are displayed with leading zeroes padded to the maximum width of the variable.

4.2.6 Define measure of data (nominal, scale etc.)

By default, variables with numeric responses are automatically detected as “Scale” variables. If the numeric responses actually represent categories, you must change the specified measurement level to the appropriate setting.

To define a variable’s measurement level, click inside the cell corresponding to the “Measure” column for that variable. Then click the dropdown arrow to select the level of measurement for that variable: Scale, Ordinal or Nominal.



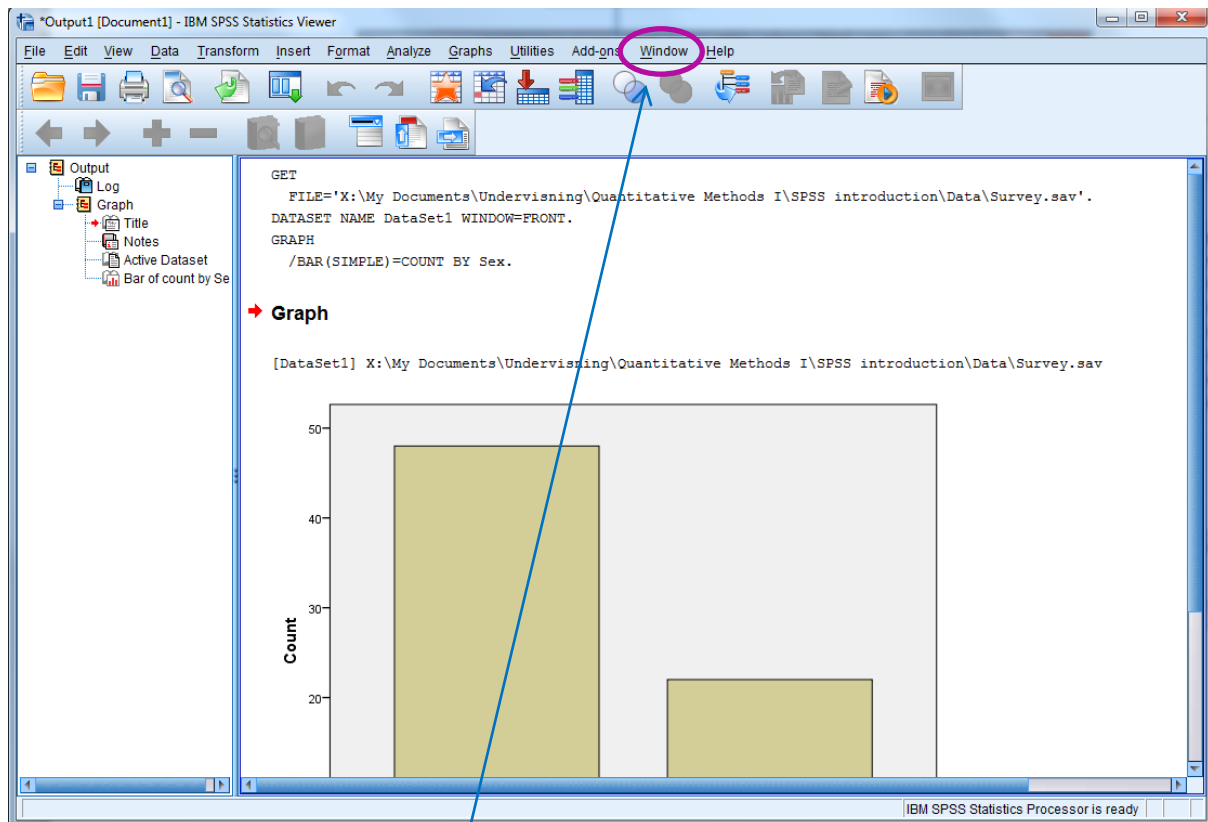
Nominal – is used for categorical data, where each value has been assigned to a discrete category. For instance, eye color of participants in a study might be nominally (from Latin *nomen* for name) categorized into groups: brown, blue, green, other.

Ordinal – the ordinal level of measure is used for data which form discrete categories and can be naturally ranked on some scale.

Scale – the scale values represents ordered categories with a meaningful metric, so that distance comparisons between values are appropriate (for example: a scale with age).

4.3 Output window

Whenever a command is carried out, a separate output window will appear.



Both windows (Data/Variable View and Output window) are open at the same time. If you want to look at the variable values and/or properties you have to go back to Data/Variable View e.g. by using the Window option in the Menu bar.

5 Way of working in SPSS

5.1 Before you start

Before you start working, make sure to make a copy of the original file. This way you always have the option to start all over again, in case you accidentally change or erase some variable and/or observations.

5.2 During work

Make it a habit to always write down the options you use (eg, "Analyze>>Descriptive statistics>> Explore", etc.). If you make a mistake, or decide to do something slightly different, you can easily go back and change.

Always check that any created or transformed variables contain the values that you intended. Section 5.6 describes how to move variables, which might be useful in this context.

It can be a good idea to check that any by SPSS created confidence intervals, test statistics or

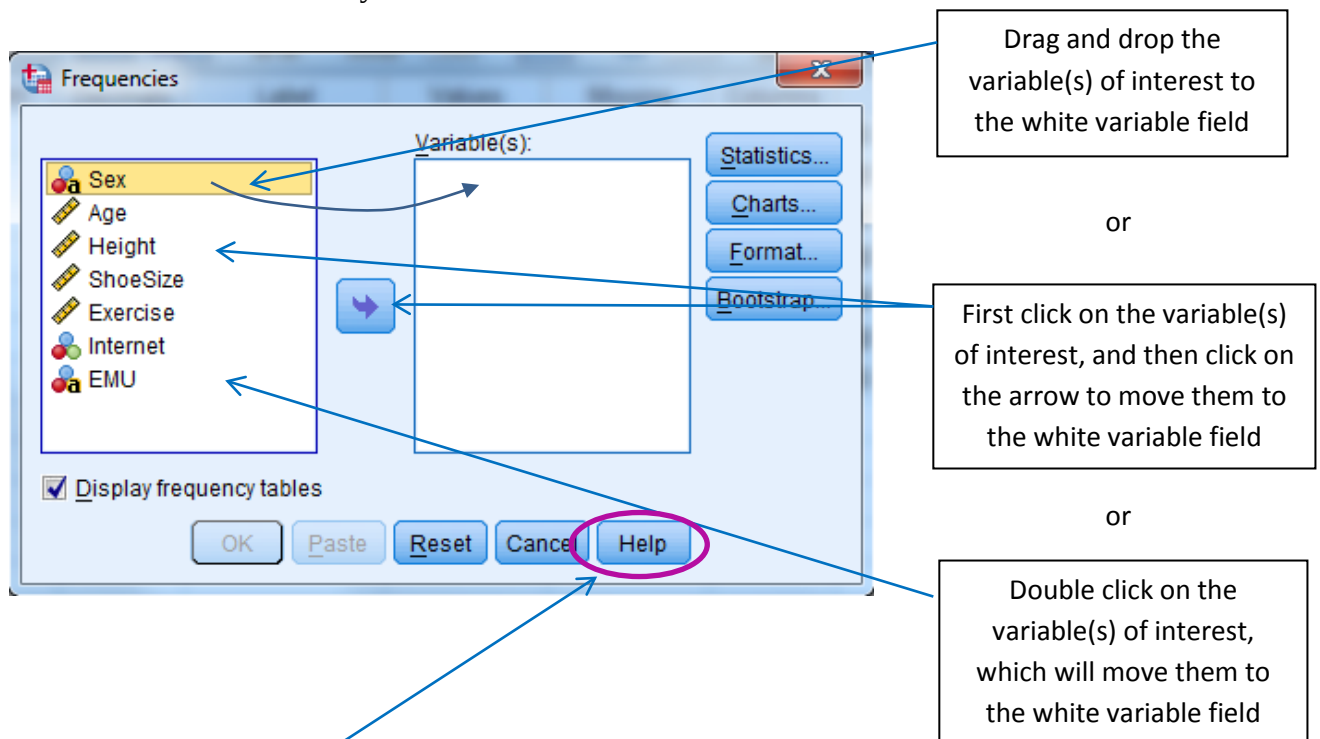
P-values are correct by calculating them manually too. At least reflect upon your results and determine whether they are reasonable or not.

5.3 Variables in columns

The calculations and analyses performed in SPSS are usually based on variable information, i.e. information in the different *columns*.

5.4 Dialog Windows

When performing statistical analyses or calculations one or several *dialog windows* often appear. In these dialog windows you have to define which variables you want to study. This can be done in different ways:

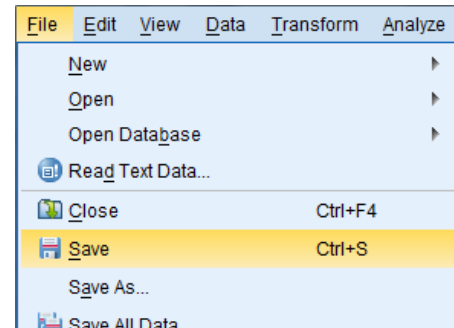


There is a Help button in every dialog window that you can use to get more information regarding the particular procedure.

5.5 Saving data and/or output

When saving your work in SPSS you can choose to save the dataset only, or also to save the output. We recommend that you save the output as well, since the output shows which analyses you have performed and this can make it easier e.g. to repeat analyses or calculations.

You can **save the dataset** by choosing **File >> Save** (or Save As) from the Menu bar in the **Data Editor window** (Data/Variable View). You will also be asked to save the data when closing SPSS.

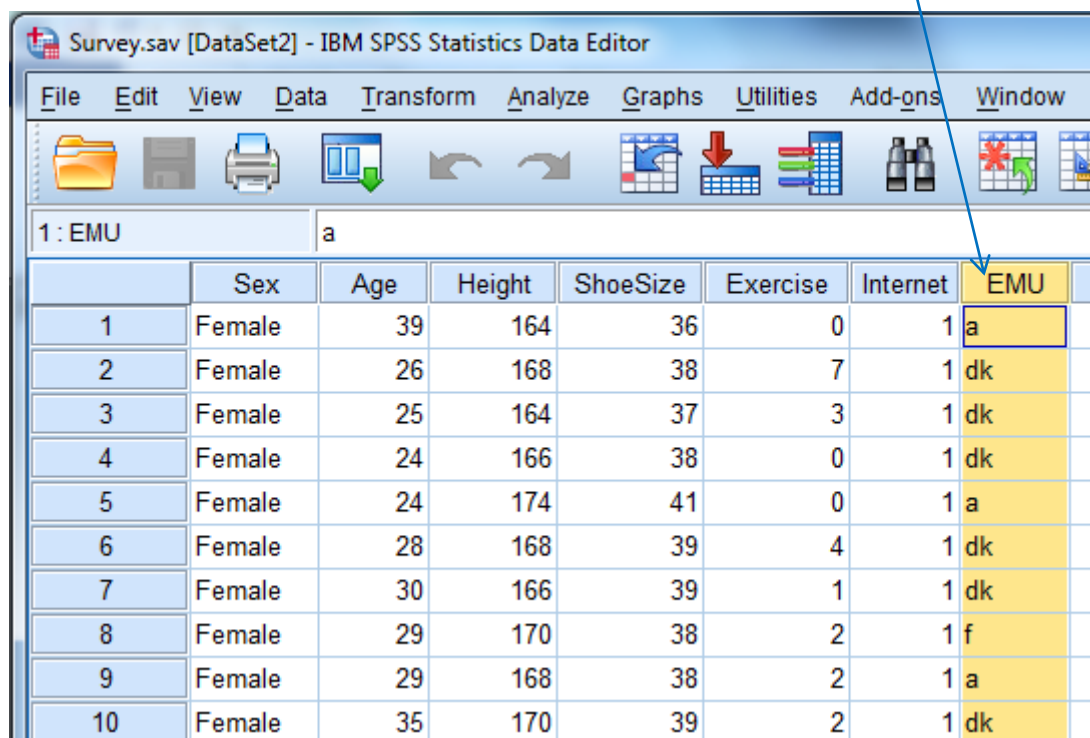


You can **save the output** by choosing **File >> Save** (or Save As) from the Menu bar in the **Output window**. You will also be asked if you want to save the output when closing SPSS.

5.6 Moving columns (variables)

When working with data you may wish to move the columns (variables), e.g. to make it easier to compare the values of two particular variables.

Select the variable you want to move by clicking on the variable's name (the column will be yellow marked), then drag-and-drop the variable to where you want it.



Moving variables is particularly useful when you create a new version of an already existing variable, e.g. creating categories from a numerical variable, or creating a numerical variable

from a text variable. By placing the two variables next to each other it is easy to compare them to check that the new variable contains what you intended.

5.7 Sorting data

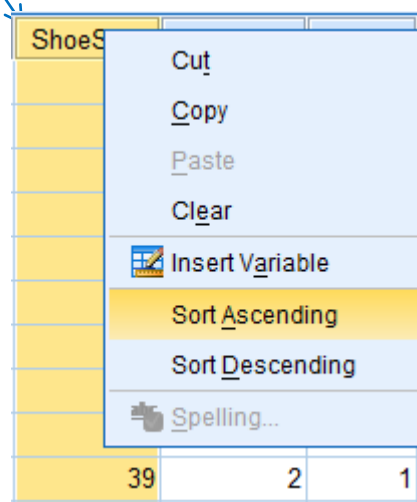
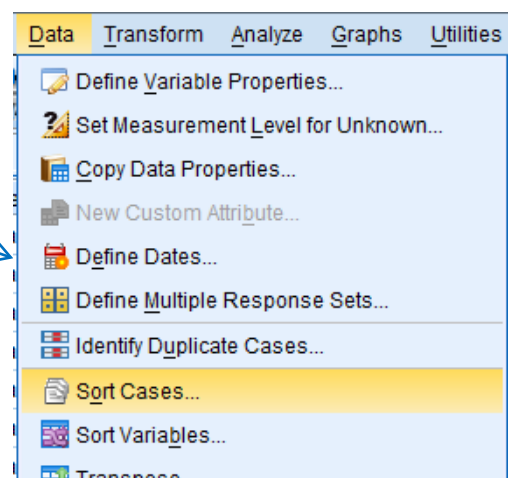
Often you want the individuals/cases in a dataset to follow a certain logical sequence. Data can be sorted ascending, with the lowest values first, or descending, with the highest values first.

You can sort the data in different ways, see below.

Use the Menu bar and choose **Data >> Sort Cases**

This option enables you to sort by more than one variable. If you sort by e.g. gender first, and then by height, you will see one gender at the top, sorted by their heights, and at the bottom you will see the other gender, sorted by their heights.
or

Right click on the variable's name and choose **Sort Ascending** or **Sort Descending**.



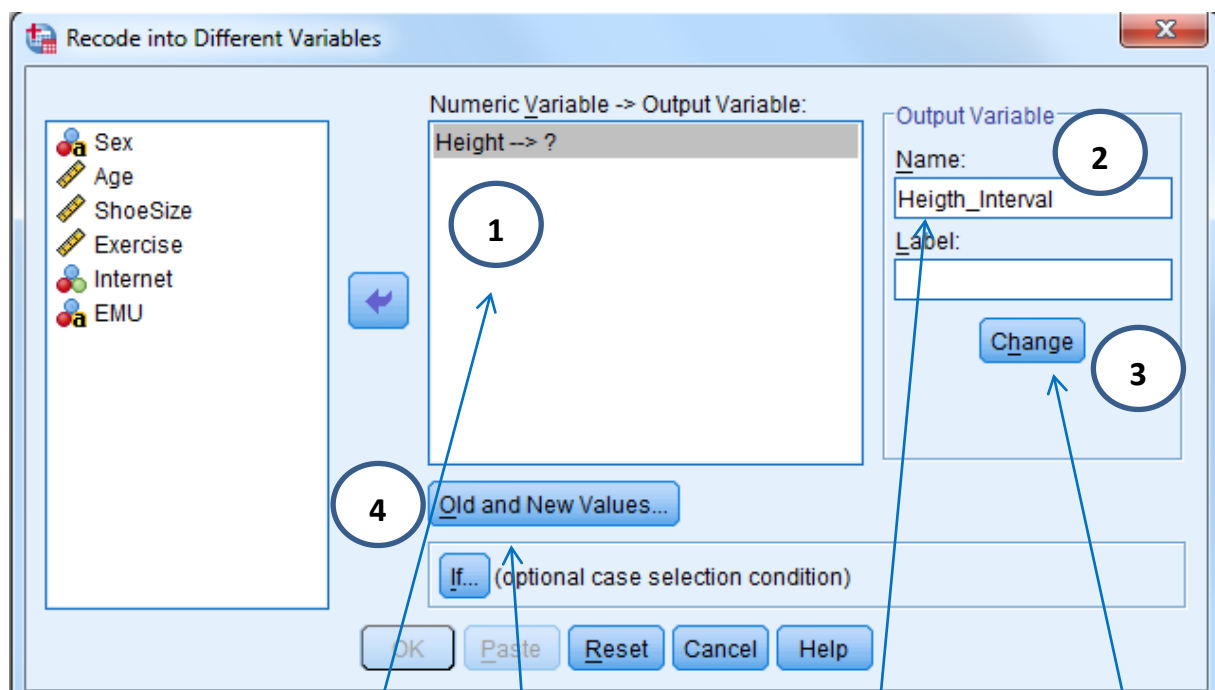
5.8 Creating new variables

Quite often you want to use existing variables to create new variables, e.g. when you want to use numbers instead of text for an ordered variable, or when you want to create classes or intervals for a continuous variable. This can be done in different ways; two ways are described in sections 5.8.1 and 5.8.2 below.

The new variable is displayed in the Data Editor. Since the variable is added to the end of the file, it is displayed in the far right column in Data View and in the last row in Variable View.

5.8.1 Recode variables (e.g. creating classes or intervals, recoding text into numerical values, or creating dummy variables)

If you want to create classes or intervals you can choose **Transform >> Recode into Different Variables** from the Menu bar. The following dialog window will appear.



1) Choose which input variable you want to use, to create the new variable from (Height is being used in this example)

2) Type a name of the variable you are about to create

3) Click "Change"

4) Click "Old and New Values"

A new dialog window will appear, see below.

1a) To e.g. recode a text variable into a numerical variable, select "Value" and type the value you want to recode.

1b) To create one of the intervals, type the range of the interval. Remember to use proper limits for the intervals/classes!

Other alternatives can be chosen here, such as individual values (not creating intervals), missing values, etc.

"Range, LOWEST through value:" creates an interval from the lowest existing value through the value that you type.

"Range, value through HIGHEST:" creates an interval from the value that you type through the highest existing value.

2) Type the value that you want the new variable to get. If you want the new variable to be a text variable, select "Output variables are strings".

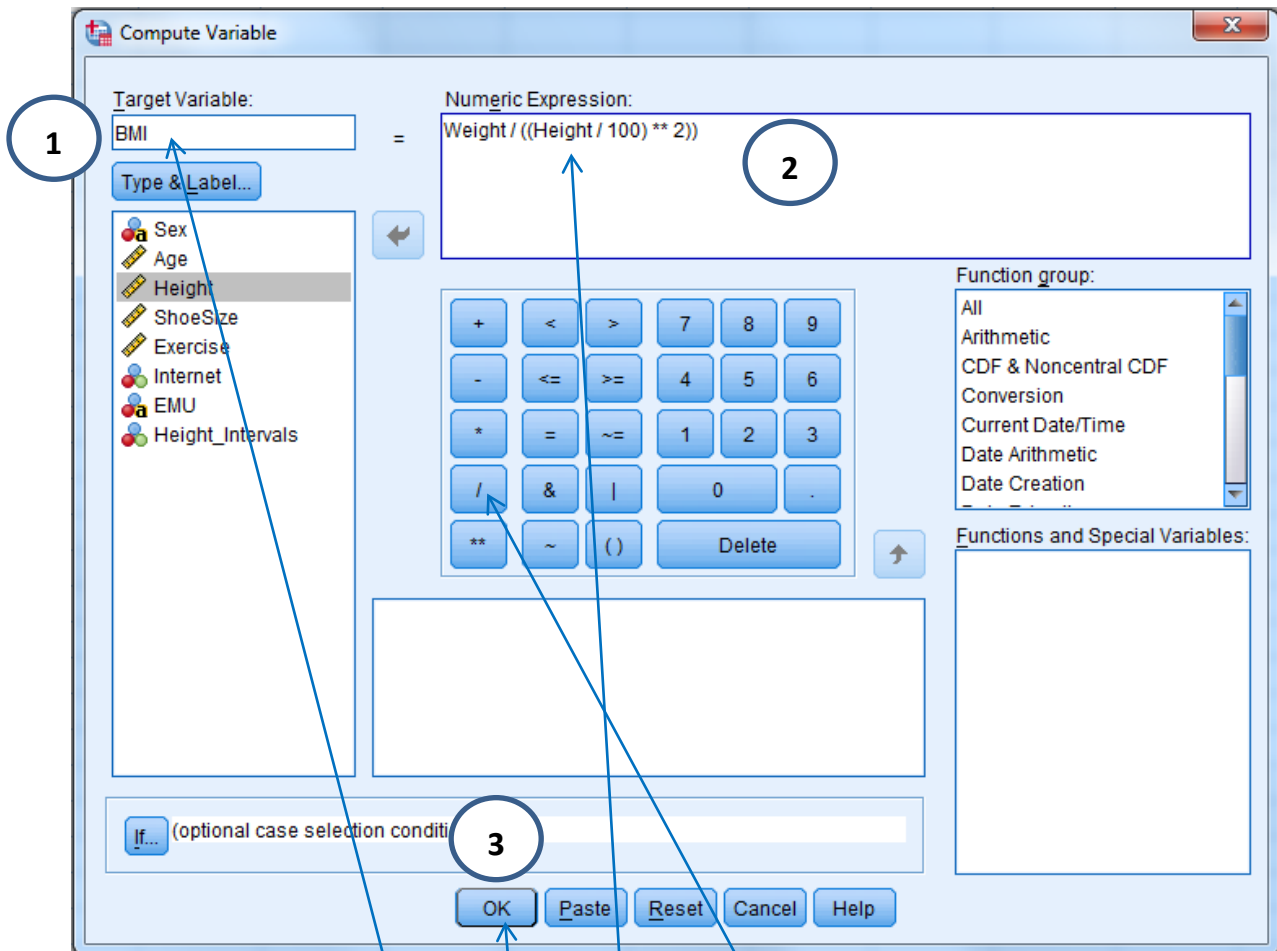
3) Click "Add"

Repeat steps 1 through 3 to add all values/intervals. Then click "Continue", and finally click "OK".

IMPORTANT! Make sure to visually check that your new variable contains the values that you intended. To make this check easier you might want to place the new variable next to the original one, see section 5.6 for how to move a column.

5.8.2 Calculate variables

Sometimes you want to create a new variable that is a function of another variable (or several other variables). Then you can choose **Transform >> Compute Variable** from the Menu bar. The dialog window below will appear.



1) Type a name of the variable you are about to create.

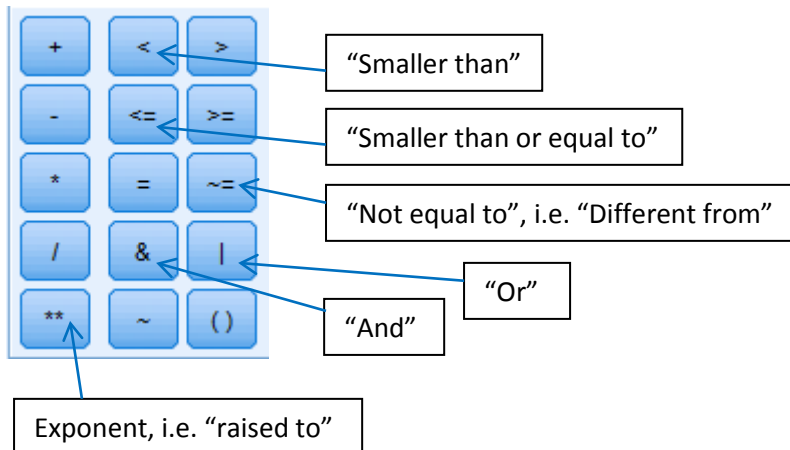
2) Type the numeric expression.

To e.g. calculate $BMI = \frac{Weight (kg)}{Height (m)^2}$, first choose variable "Weight" among the existing variables. Then type a division mark ("slash"), or click on the corresponding blue button (see an explanation of the most common expressions in section 5.8.3 below), and complete the expression.

3) Click "Change"

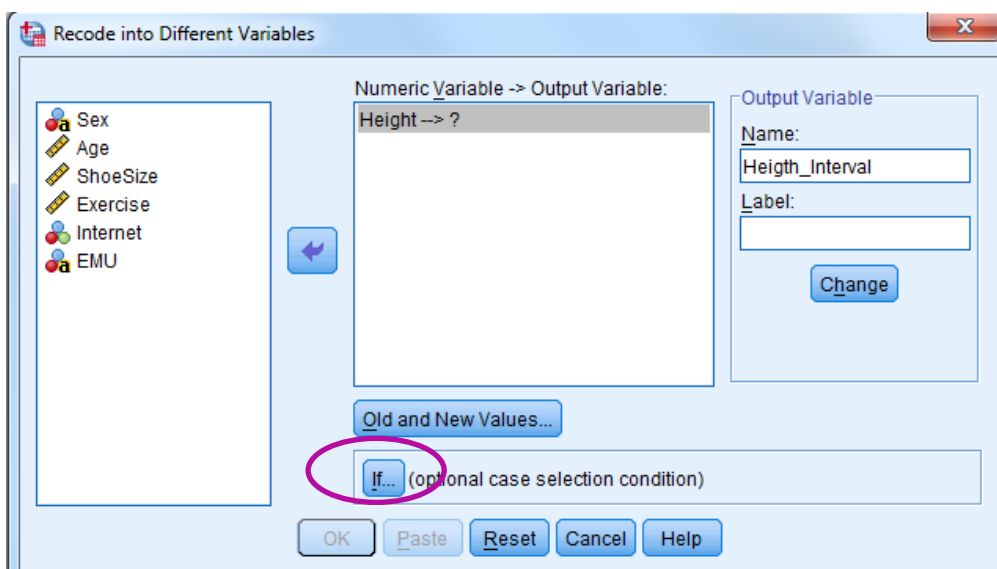
5.8.3 Numerical operands

When creating new variables you may want to use mathematical expressions of different kinds. It is very important to use correct expressions, and SPSS has some built in numerical operands that can be used. The most common ones are explained below.

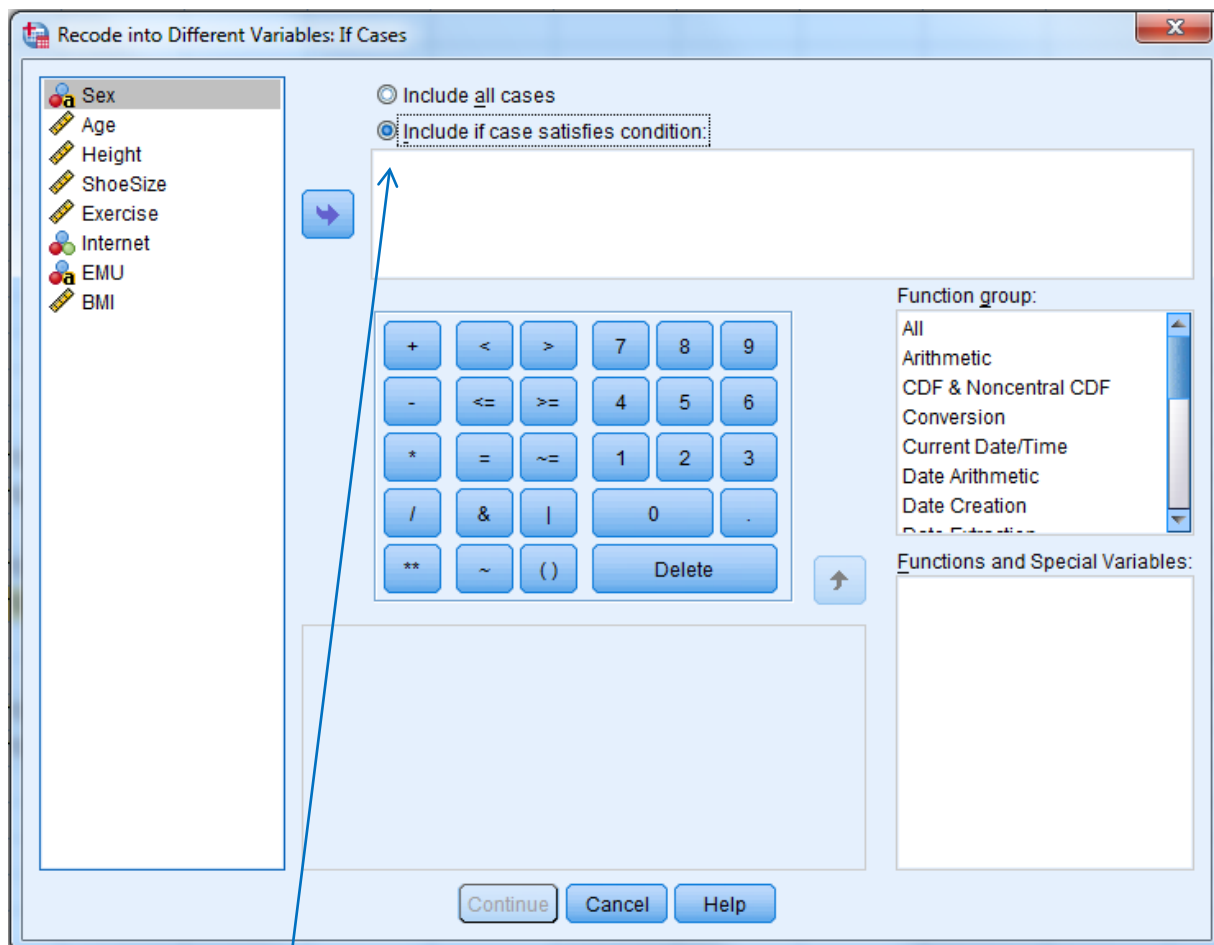


5.8.4 The "if" function

When creating new variables, either by recoding or calculating them, you might want to perform the particular operation only for a subset of the individuals (e.g. for males only). Then you can use the "if" function.



Click "If...", and the dialog window below will appear.



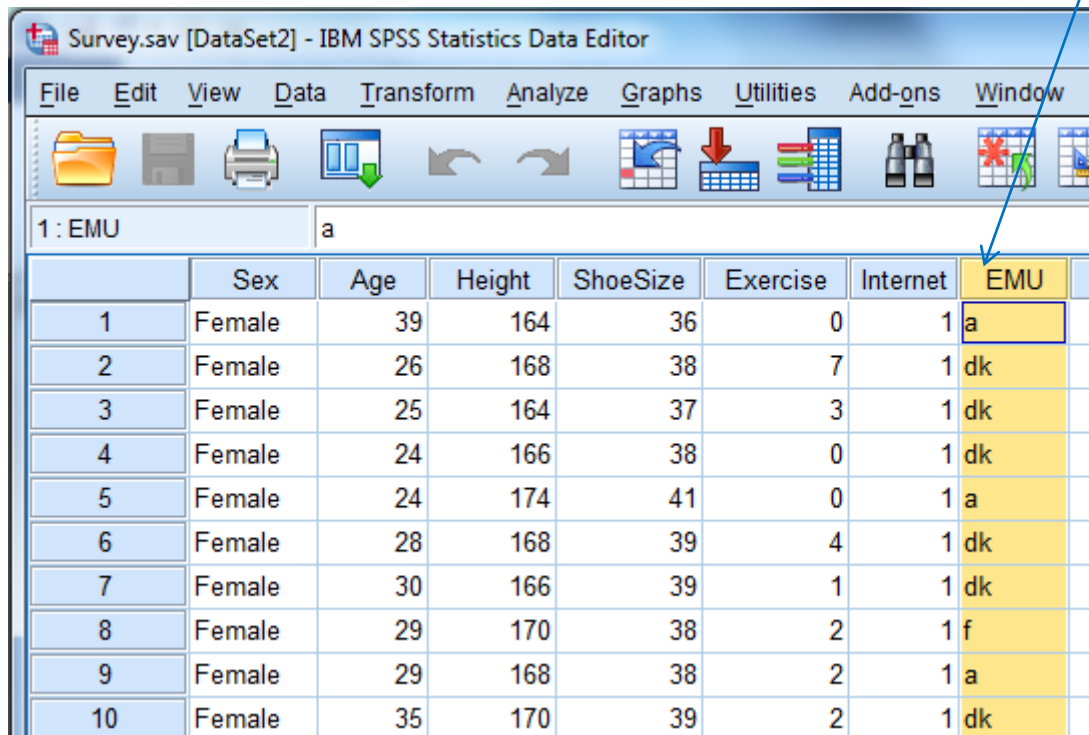
Mark "Include if case satisfies condition:"
and type the condition (numeric expression) in the white field.
Examples of numeric expressions are presented in section 5.8.5 below.

5.8.5 Examples of numeric expressions

Expression	Result
Sex = 'Female'	Will perform the operation for females only
(Height >=160) & (Height <=180)	Will perform the operation only for individuals that are 160 to 180 cm tall
ShoeSize > 39	Will perform the operation only for people with Shoe size larger than 39
(Exercise >=1) (Age <= 40)	Will perform the operation for people who exercised at least once the past week, or being of the age of 40 years or less

5.9 Deleting variables (columns)

The easiest way to delete a variable is to select the variable by clicking on the variable name (the column will then be yellow marked) and then press Delete.

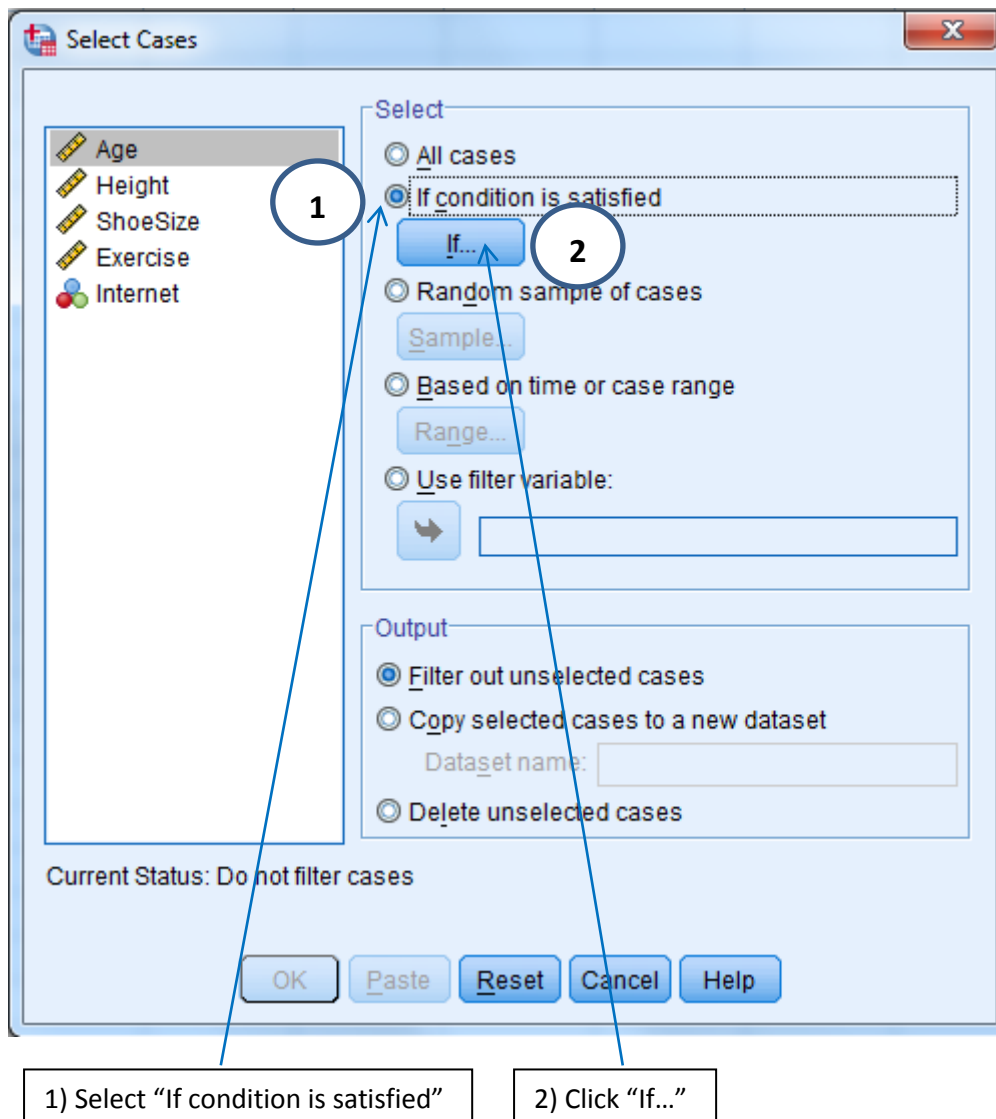


5.10 Make calculations for selected individuals

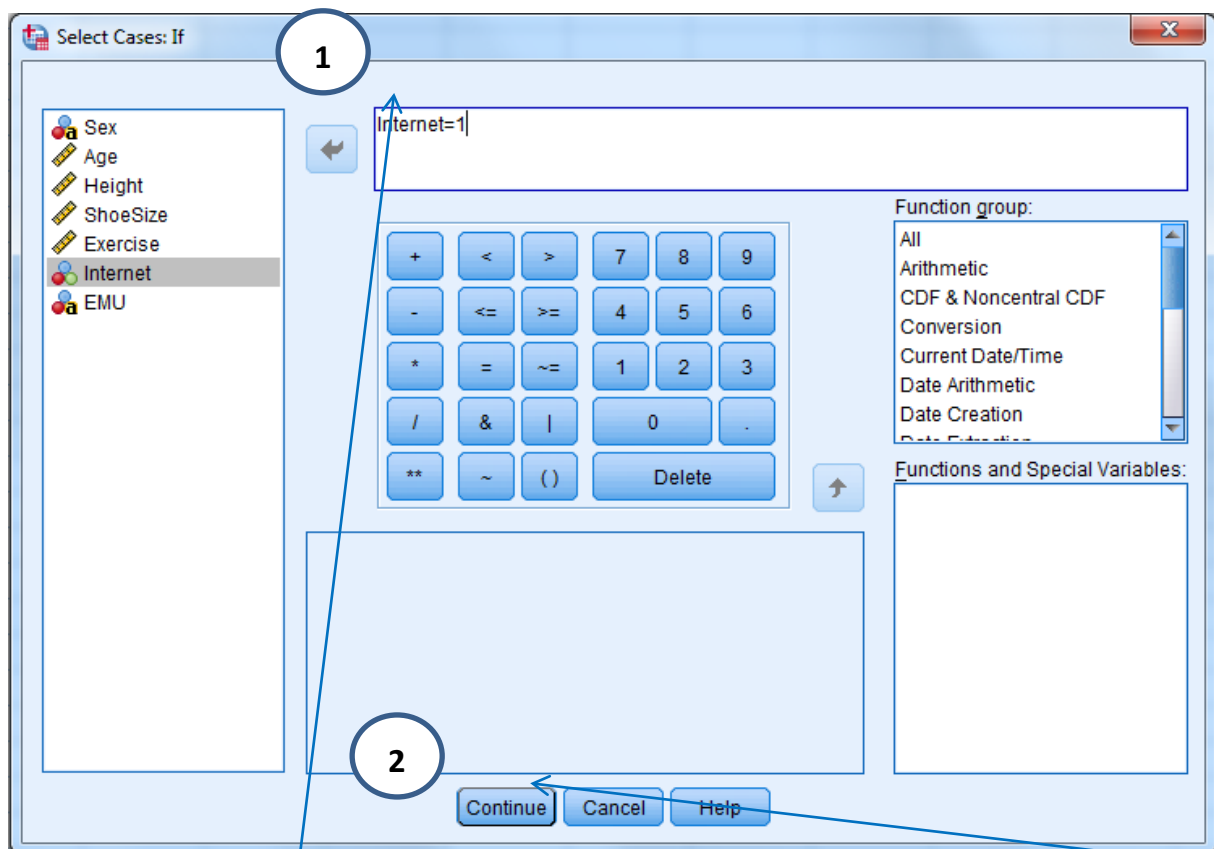
Sometimes you want to perform the analyses only for a certain number of individuals/cases, or you want to perform the analyses separately for different groups (e.g. by gender). Section 5.10.1 below describes how to make a subset of data, and section 5.10.2 describes how to split the data into groups.

5.10.1 Make a subset of data (Select Cases)

Choose **Data >> Select Cases** from the Menu bar. This will open the dialog window below.



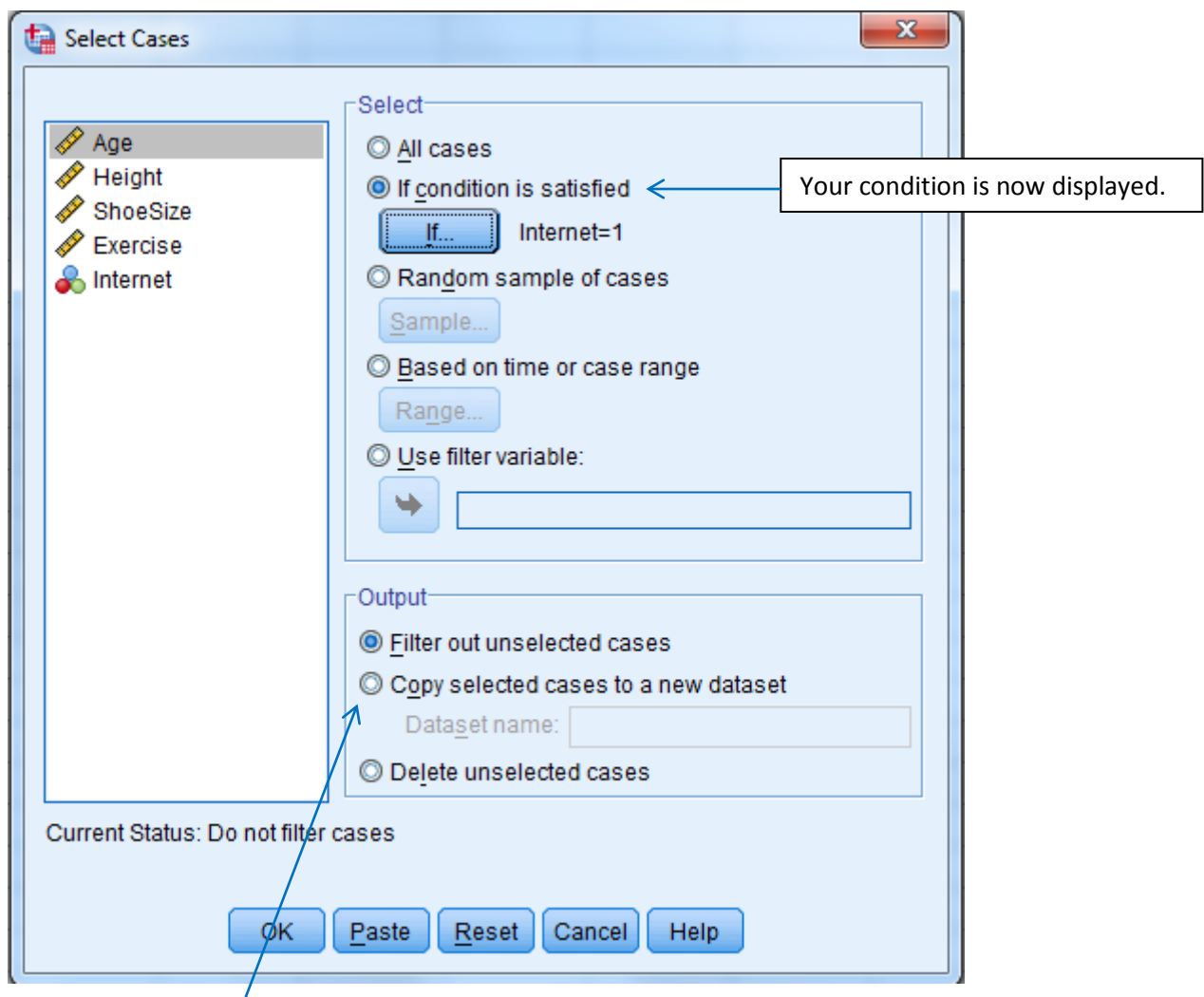
The dialog window below will appear.



- 1) Type the condition (numeric expression) in the white field.
Individuals who fulfill the condition will be selected (in this example only individuals with internet connection at home will be selected).
Examples of other numeric expressions are presented in section 5.8.5 above.

- 2) Click
"Continue"

This will bring you back to the first dialog window (see below).



Choose what you want to do with the selected cases/individuals.

Filter out unselected cases: A new variable named filter_\$ will be created, where the value 1 denotes selected cases/individuals and 0 denotes unselected cases/individuals. Unselected cases will be marked in the Data Editor with a diagonal line through the row number.

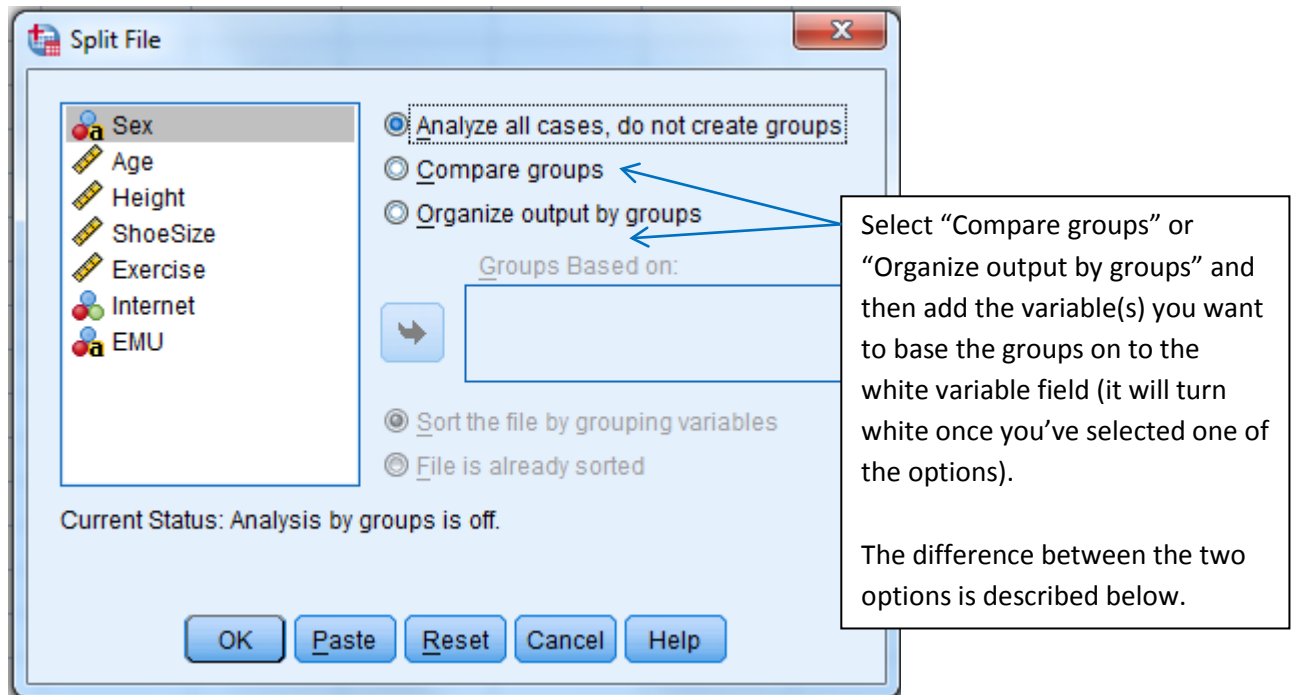
Copy selected cases to a new dataset: A new dataset will be created, which contains only selected cases/individuals .

Delete unselected cases: Selected cases/individuals will be deleted.

IMPORTANT! Make sure to save the dataset under a new name! Deleted cases can be recovered only by exiting from the file without saving any changes and then reopening the file. The deletion of cases is permanent if you save the changes to the data file.

5.10.2 Split the data into separate groups (Split-File processing)

To split your data file into separate groups for analysis, choose **Data >> Split File** from the Menu bar. The following dialog window will appear.



Compare groups: Results from all split-file groups will be included in the same table(s).

Organize output by groups: The file is split into separate groups for the chosen variable(s), and all output will be provided separately for each group.

NOTE! After you invoke split-file processing, it remains in effect for the rest of the session unless you turn it off. To turn it off, choose **Data >> Split File** from the Menu bar again and select "Analyze all cases, do not create groups".

6 Entering data manually

To enter data manually, follow the instructions in **chapter 3** of **IBM SPSS Statistics 22 Brief Guide**:

ftp://ftp.software.ibm.com/software/analytics/spss/documentation/statistics/22.0/en/client/Manuals/IBM_SPSS_Statistics_Brief_Guide.pdf

With these instructions you'll learn how to:

- Enter numeric or string data with variables as columns, individuals as rows
- Define the variables by using Variable View
- Define variable types (numeric or string)
- Add variable labels (descriptions of the variables)
- Change variable type and format
- Add value labels (1="male", etc.)
- Handle missing values

7 Reading data in from Excel

7.1 A couple of warnings

IMPORTANT! Before you import data from Excel, you have to make sure that there aren't any missing values in the Excel file. Missing values in Excel will be coded as zeroes (0) in SPSS! Instead, code any missing values using a number that your variable cannot take and that will be easy to spot, e.g. 99999.

Ensure that the columns represent variables, and rows represent individuals.

Also make sure that any variable names are contained in the first row only in the Excel file, and that the variable values start at the second row.

7.2 Importing the data into SPSS

Data can then be imported to SPSS from Excel the following way:

- 1) Open SPSS.
- 2) If you are prompted with a window that asks "What would you like to do?" choose the second option, "Type in data."
- 3) Choose **File >> Open >> Data** from the Menu bar.
- 4) Choose "Files of type: Excel", and then click the Excel file you want to import and click "Open".
- 5) If your Excel file contains multiple worksheets, select the worksheet you want to import.
- 6) Click OK, and the data set is imported.

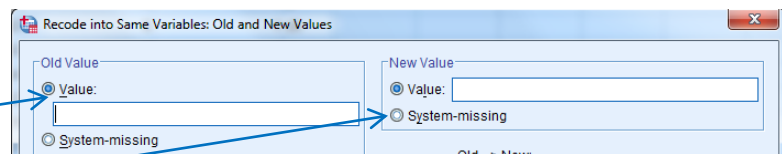
7.3 Missing values

Remember to let SPSS know which value(s) that denote missing values.

This can be done in two different ways.

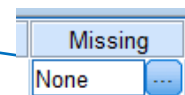
7.3.1 Deleting missing values (numerical variables only)

- 1) Choose **Transform >> Recode into Same Variables** from the Menu bar.
- 2) Add the variable(s) for which you want to delete the missing values, by e.g. clicking and pulling them to the “Numerical variables” field. You can add all the variables for which you’ve used the same missing code.
- 3) Click “Old and New Values”
- 4) To the left, under “Old value”, mark “Value” and type the missing code (e.g. 9999) in the white box.
- 5) To the right, under “New value”, mark “System-missing”
- 6) Click “Add”
- 7) Click “Continue”
- 8) Click OK



7.3.2 Coding missing values (e.g. for string variables)

- 1) Click on the Variable View tab in the bottom left hand corner of the data editor window.
- 2) Look at the row for the variable you’re dealing with and go to the Missing column.
- 3) Click on the word None.
- 4) Click on the little grey square (with dots in it) on the right.
- 5) Mark “Discrete missing values” and type the number you’ve chosen to denote missing values for this variable in the first white box.
- 6) Click OK.
- 7) Repeat until you have entered all the missing codes for all variables.



8 Using data in other formats

8.1 Reading data in from text files

To import data text files, follow the instructions in **IBM SPSS Statistics 22 Brief Guide pp. 12-14:**

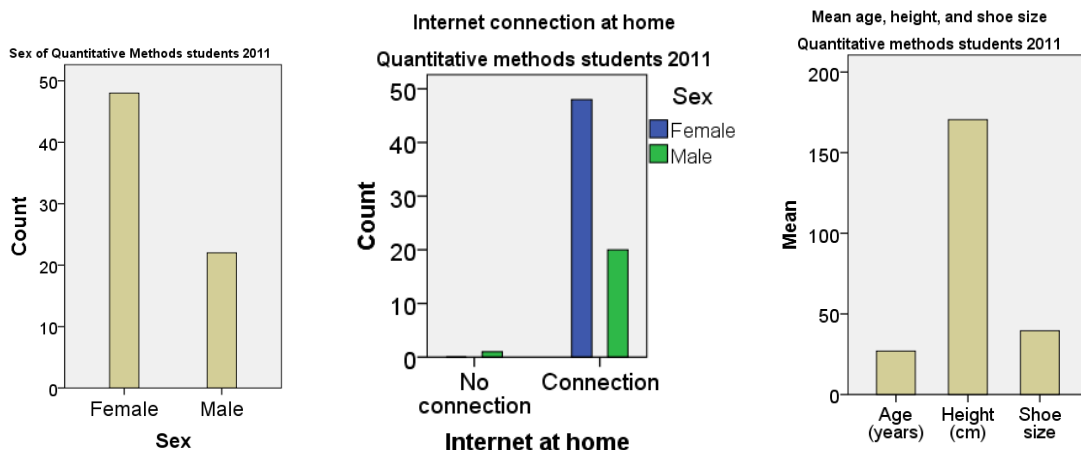
[ftp://ftp.software.ibm.com/software/analytics/spss/documentation/statistics/22.0/en/client/Manuals/IBM SPSS Statistics Brief Guide.pdf](ftp://ftp.software.ibm.com/software/analytics/spss/documentation/statistics/22.0/en/client/Manuals/IBM_SPSS_Statistics_Brief_Guide.pdf)

8.2 Using the data sets on the CD

- 1) Open SPSS.
- 2) If you are prompted with a window that asks "What would you like to do?" choose the second option, "Type in data."
- 3) Choose **File >> Open >> Data** from the Menu bar.
- 4) Choose "Files of type: Portable (*.por)"
- 5) Click the SPSS file you want to import and click "Open".
- 6) Click OK, and the data set is imported.

9 Creating graphs

9.1 Bar charts



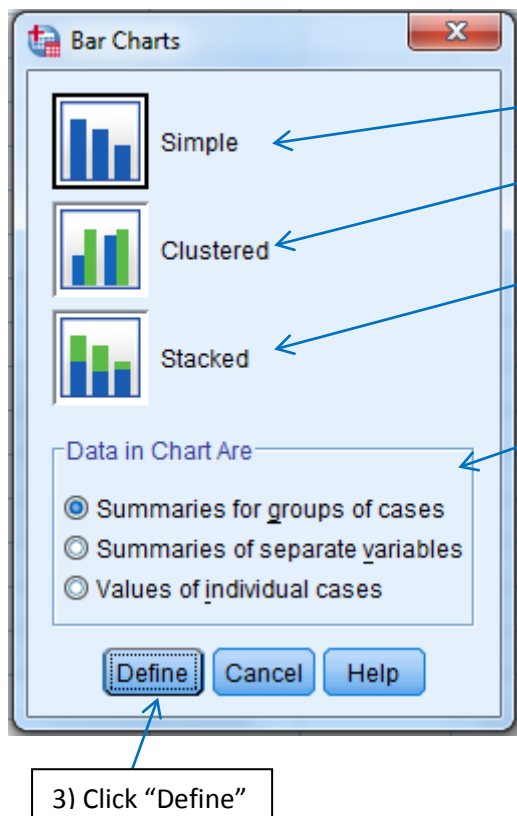
Simple bar chart

Clustered/grouped bar chart

Simple bar chart, Summaries of separate variables

The distribution of a categorical (qualitative) variable can be visualized by a bar chart.

Choose **Graphs >> Legacy Dialogs >> Bar** from the Menu tab. The dialog window below will appear.



1) Click the type of bar chart you wish to produce.

Simple: Displays one variable.

Clustered: Displays one variable, **grouped** by a second variable.

Stacked: Displays one variable, stacked by a second variable.

2) Choose what you want the graph to contain.

Summaries for groups of cases: Displays the categories of one variable.

Summaries of separate variables: Displays the mean (other measures can also be chosen) for one or several variables.

Values of individual cases: Displays one bar for each individual.

9.1.1 Simple bar charts

If you choose "Simple" above, the dialog window below will appear.

1) Choose what you want the bars to represent
“N of cases” = number of cases/individuals in each category
“% of cases” = percent of cases/individuals in each category

The screenshot shows the 'Define Simple Bar: Summaries for Groups of Cases' dialog box. On the left is a list of variables: Nr, Age (years) [Age], Height (cm) [Height], Shoe Size [ShoeSize], Exercise (number of...), Internet, EMU, EMU_new, Internet connection [...], EMU preference [EM...], Sex_N, and EMU preference [E...]. The 'Bars Represent' section has four radio buttons: 'N of cases' (selected and circled with 1), 'Cum. N', '% of cases' (circled with 2), and 'Cum. %'. Below this is a 'Variable:' field with an arrow button and a 'Change Statistic...' button. The 'Category Axis:' section has a list box containing 'Sex' (circled with 3) and an arrow button. Below this is the 'Panel by' section with 'Rows:' and 'Columns:' list boxes, each with an arrow button, and checkboxes for 'Nest variables (no empty rows)' and 'Nest variables (no empty columns)'. On the right side of the dialog are 'Titles...' and 'Options...' buttons (circled with 4). At the bottom is a 'Template' section with a checkbox 'Use chart specifications from:' and a 'File...' button. At the very bottom are buttons for 'OK' (circled with 5), 'Paste', 'Reset', 'Cancel', and 'Help'. Arrows point from the numbered text boxes to these specific elements in the dialog.

3) Click “Titles”, and type an informative title explaining what the graph is displaying.

4) Click “Options”, and select “Display groups defined by missing values” if you want to include a bar to represent missing values.

2) Add your variable of interest to the “Category Axis” field

5) Click “OK” to produce the graph

9.1.2 Clustered (grouped) bar charts

If you choose “Clustered” in the first bar chart dialog window (see section 9.1 above), the following dialog window will appear.

Define Clustered Bar: Summaries for Groups of Cases

1) Choose what you want the bars to represent. Percent of cases/individuals in each category is often the most appropriate alternative when comparing different groups.

2) Add your variable of interest to the “Category Axis” field

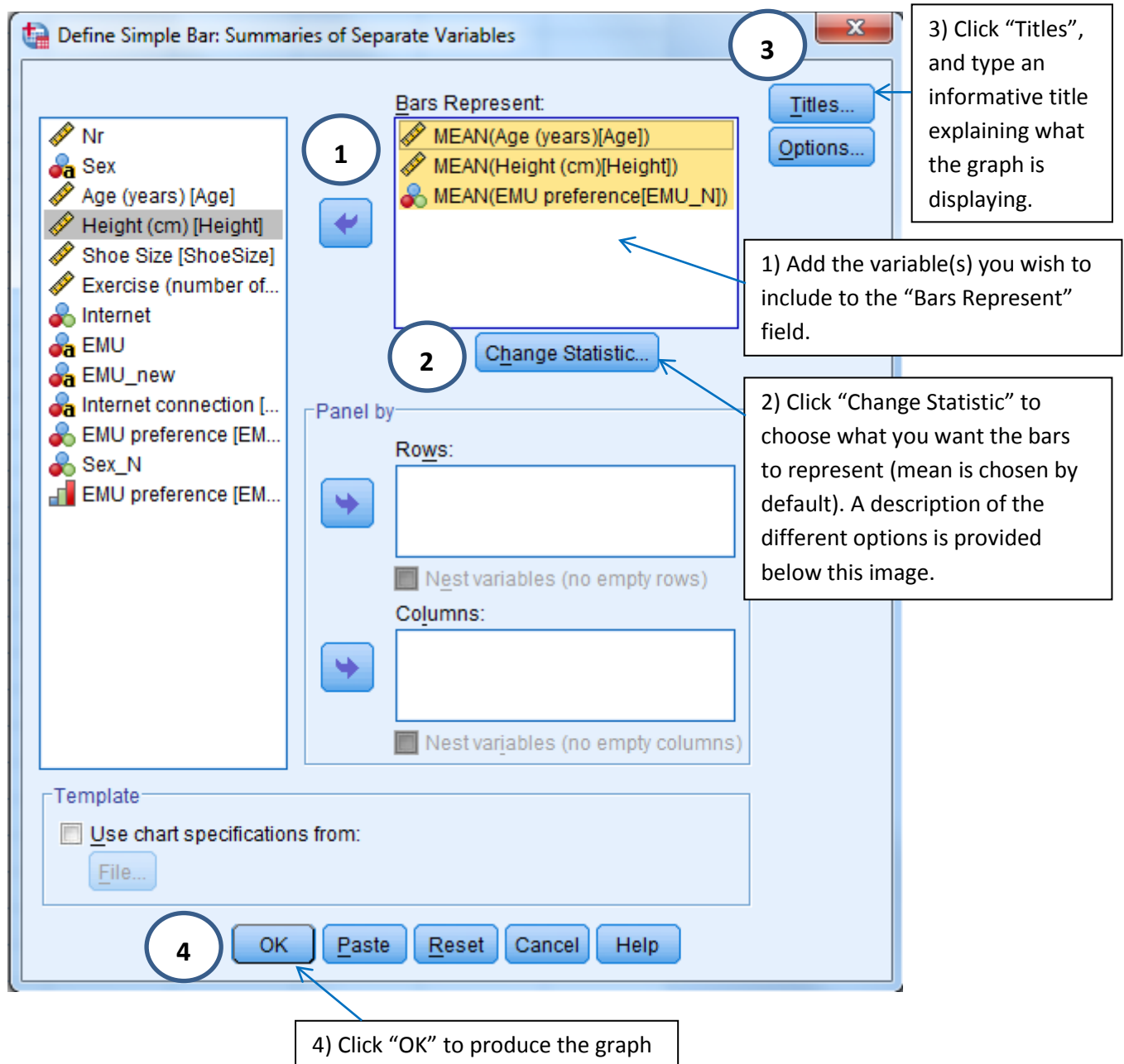
3) Add the grouping variable to the “Define Clusters by” field

4) Click “Titles”, and type an informative title explaining what the graph is displaying.

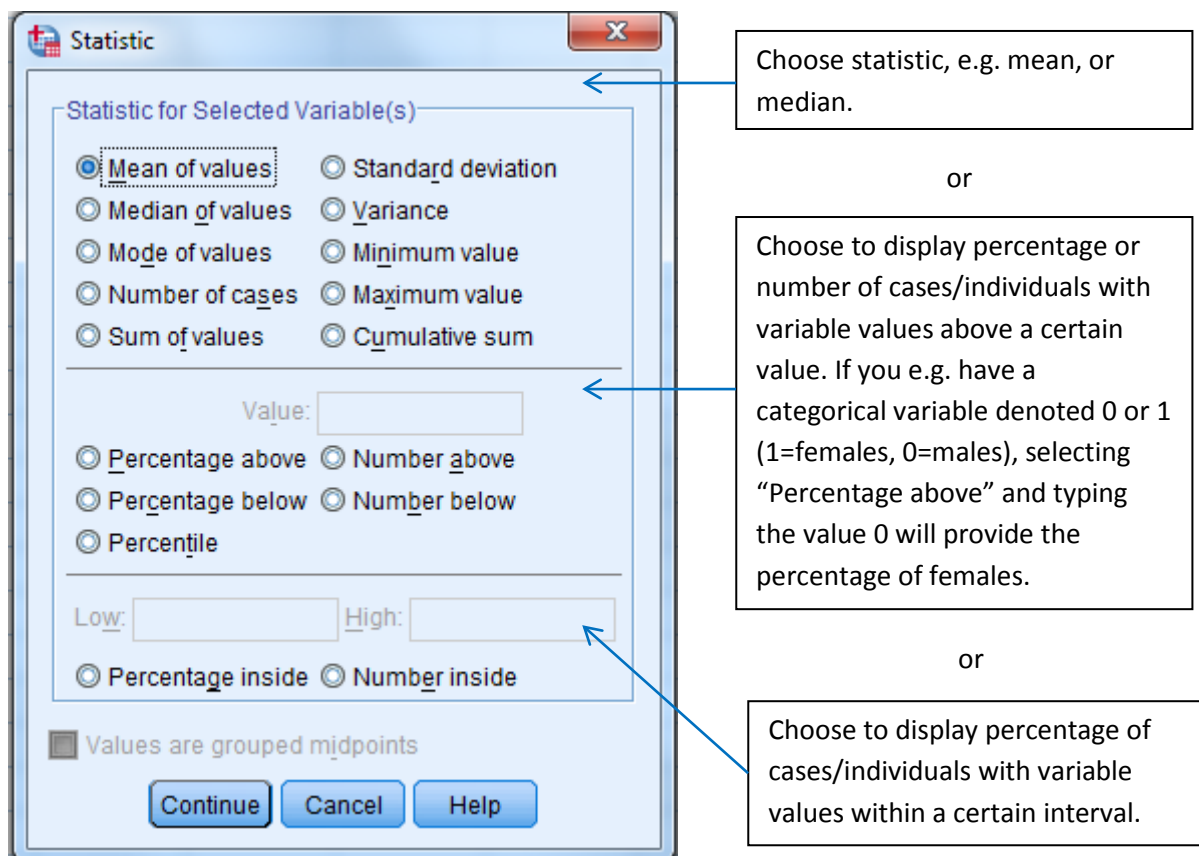
5) Click “OK” to produce the graph

9.1.3 Several variables in the same bar chart

To include several variables in the same bar chart, choose “Summaries of separate variables” in the first bar chart dialog window (see section 9.1 above). The following dialog window will then appear.



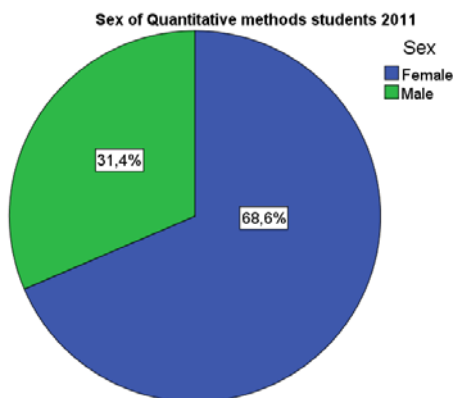
If you click “Change Statistic” above, the dialog window below will appear.



Then click "Continue" to get back to the previous dialog window.

And finally, click "OK" to produce the graph.

9.2 Pie charts

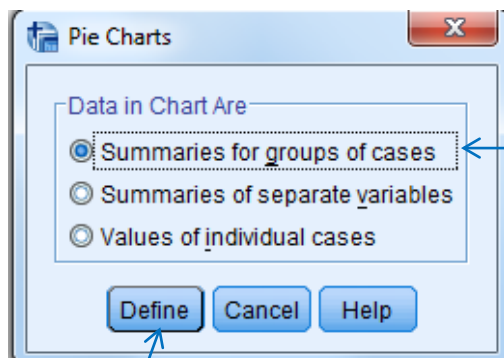


Pie chart

Summaries for groups of cases

The distribution of a categorical (qualitative) variable with few categories can be visualized by a pie chart.

Choose **Graphs >> Legacy Dialogs >> Pie** from the Menu tab. The following dialog window will appear.



2) Click "Define"

1) Choose what you want the pie chart to contain.

Summaries for groups of cases: is the most common choice. Each pie sector represents a category of the variable.

Summaries of separate variables: displays the sum of each variable's values as pie sectors (for a number of variables).

Values of individual cases: displays one pie sector for each individual/case.

If you select "Summaries for groups of cases", the dialog window below will appear.

1) Add the variable you want to produce the pie chart for to the "Define Slices" field.

2) Select what you want the pie sectors to represent
N of cases: number of cases/individuals in each category.
% of cases: percent of cases/individuals in each category.

3) Click "Titles", and type an informative title explaining what the graph is displaying.

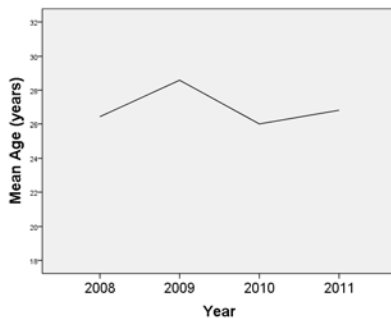
3) Click "Options", and select "Display groups defined by missing values" if you want to include a pie sector to represent missing values.

The screenshot shows the 'Define Pie: Summaries for Groups of Cases' dialog box. On the left, a list of variables includes Age, Height, ShoeSize, Exercise, Internet, and EMU. In the center, the 'Slices Represent' section has 'N of cases' selected. Below it, the 'Define Slices by:' section has 'Sex' selected. The 'Panel by' section has empty fields for 'Rows' and 'Columns'. At the bottom, the 'Template' section has 'Use chart specifications from:' checked. Numbered circles 1 through 5 point to specific elements: 1 points to the 'Define Slices by:' field, 2 points to the 'Slices Represent' section, 3 points to the 'Titles...' button, 4 points to the 'Options...' button, and 5 points to the 'OK' button.

5) Click "OK" to produce the pie chart.

It is very informative to add counts or percentages to the pie sectors, see section 10.6 10.6.

9.3 Time plots (line charts)



Time plot

The distribution of a numerical variable over a number of categories representing points of time can be visualized by a time plot.

Choose **Graphs >> Legacy Dialogs >> Line** from the Menu tab. The following dialog window will appear.

1) Click the type of line chart you wish to produce.

Simple: Displays one variable, over different categories (points of time).

Multiple: Displays one variable, with two separate lines denoting two different groups.

Drop-line: Displays one variable, with two separate symbols denoting two different groups. The two groups are connected by a line at each time point.

2) Choose what you want the graph to contain.

Summaries for groups of cases: Displays one variable.

Summaries of separate variables: Displays the mean (other measures can also be chosen) for one or several variables.

Values of individual cases: Displays one line for each individual.

3) Click "Define"

If you choose “Simple” and “Summaries for groups of cases” above, the following dialog window will appear.

1) Add the variable you want to use as time variable to the “Category axis” field.

2) Select what you want the line to represent

3) Click “Titles”, and type an informative title explaining what the graph is displaying.

4) Click “OK” to produce the time plot (line chart).

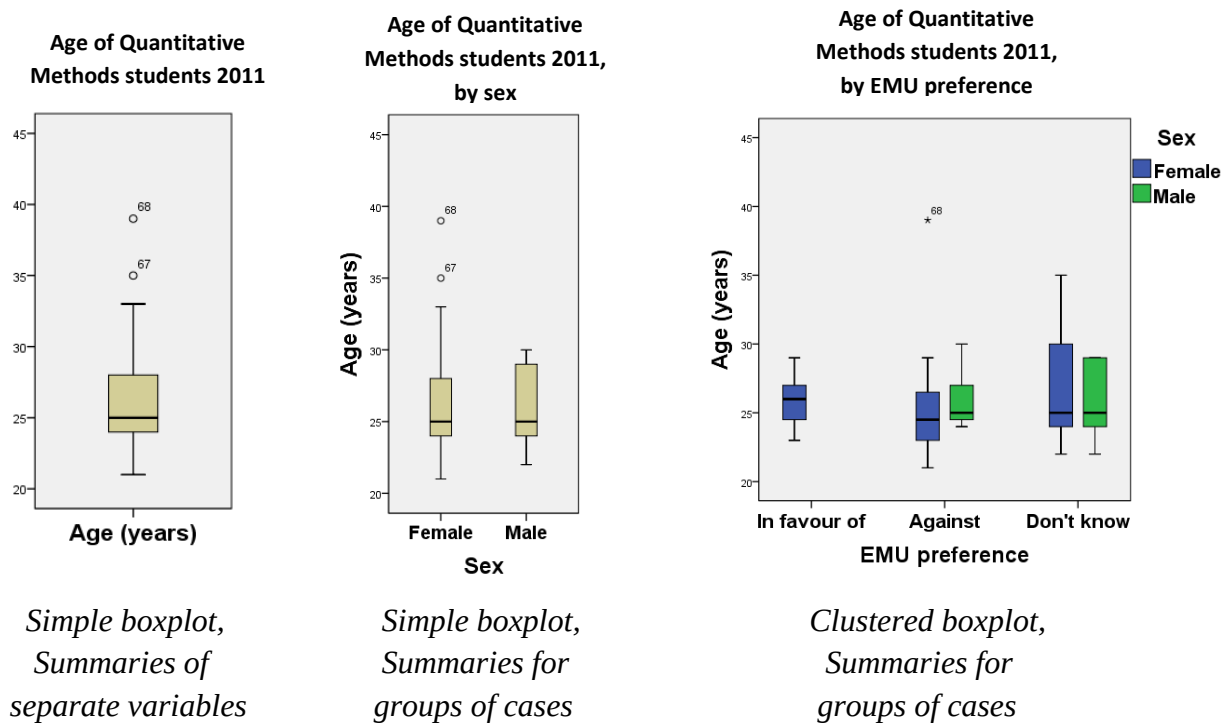
The screenshot shows the 'Define Simple Line: Summaries for Groups of Cases' dialog box. On the left is a list of variables: Sex, Height, ShoeSize, Exercise, Internet, and EMU. The 'Line Represents' section has three radio buttons: 'N of cases', '% of cases', and 'Other statistic (e.g., mean)'. The 'Variable' field contains 'MEAN([Age])'. The 'Category Axis' field contains 'Year'. The 'Panel by' section has 'Rows' and 'Columns' fields, both empty, with checkboxes for 'Nest variables (no empty rows)' and 'Nest variables (no empty columns)'. The 'Template' section has a checkbox for 'Use chart specifications from:' and a 'File...' button. At the bottom are buttons for 'OK', 'Paste', 'Reset', 'Cancel', and 'Help'. Numbered circles and arrows point to specific elements: 1 points to the 'Category Axis' field, 2 points to the 'Other statistic' radio button, 3 points to the 'Titles...' button, and 4 points to the 'OK' button.

N of cases: number of cases/individuals at each time point (in each category).
% of cases: percent of cases/individuals at each time point (in each category).
Other statistic: displays e.g. the mean of a certain variable. If you select this option, you have to add a variable to the “Variable” field.

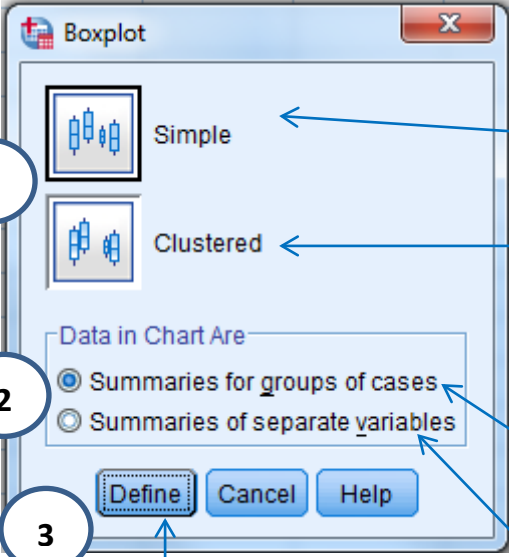
You might want to adjust the axis of the graph, see section 10.4.

9.4 Boxplots

The distribution of a numerical/quantitative variable, or an ordered categorical variable, can be visualized by a boxplot.



Choose **Graphs >> Legacy Dialogs >> Boxplot** from the Menu tab. The following dialog window will appear.



The image shows the SPSS Boxplot dialog box with three numbered annotations. Annotation 1 points to the 'Simple' and 'Clustered' boxplot icons. Annotation 2 points to the 'Summaries for groups of cases' and 'Summaries of separate variables' radio buttons. Annotation 3 points to the 'Define' button.

1) Click the type of boxplot you wish to produce.

Simple: Displays one variable, can be grouped by a second (categorical) variable.

Clustered: Displays one variable, clustered by a second variable. Can also be grouped by a third (categorical) variable.

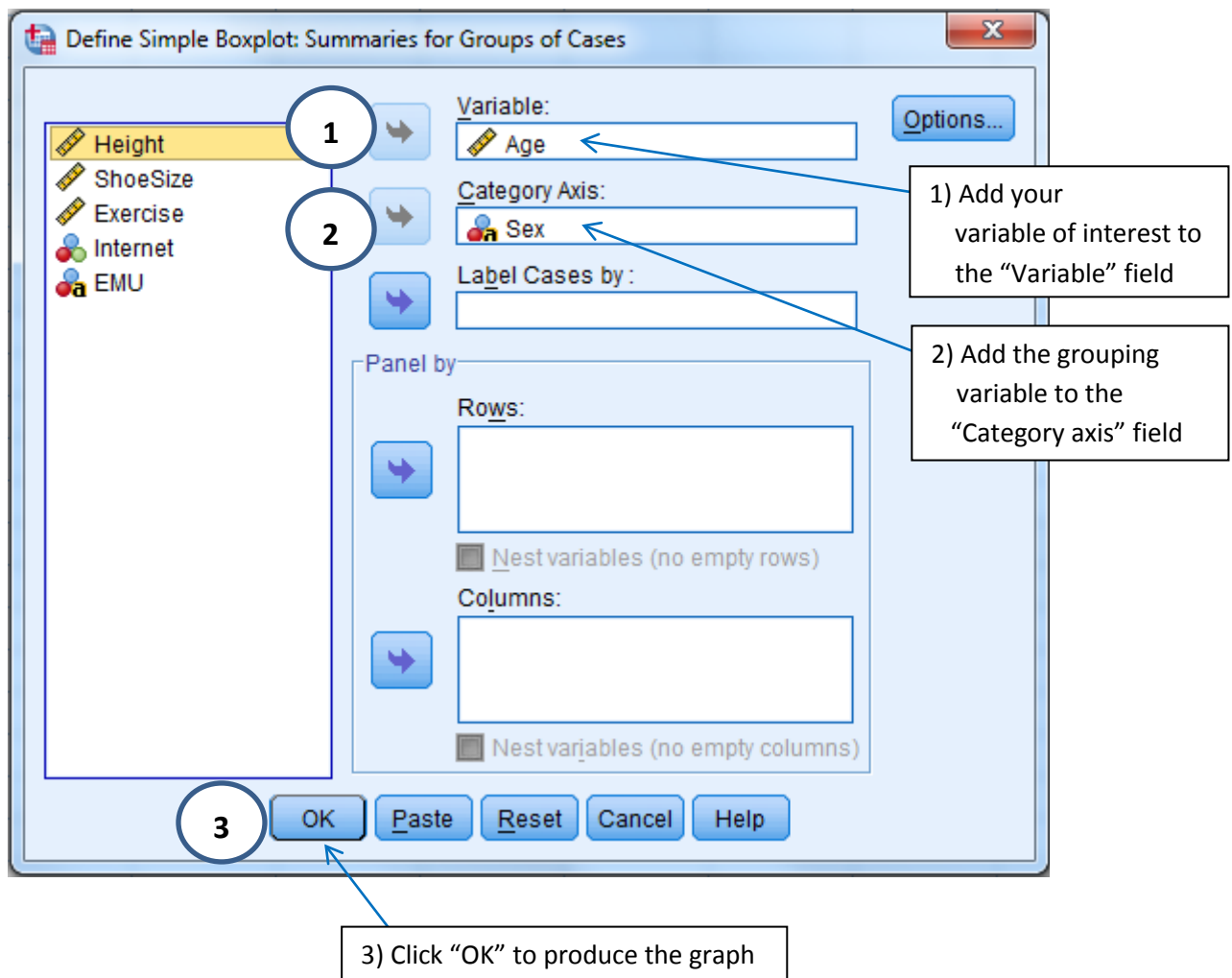
2) Choose what you want the graph to display

Summaries for groups of cases: Displays one variable, grouped by a second (categorical) variable.

Summaries of separate variables: Displays one or several variables, without grouping.

3) Click "Define"

If you choose "Simple" above (which is the most common choice), and "Summaries for groups of cases", the dialog window below will appear.

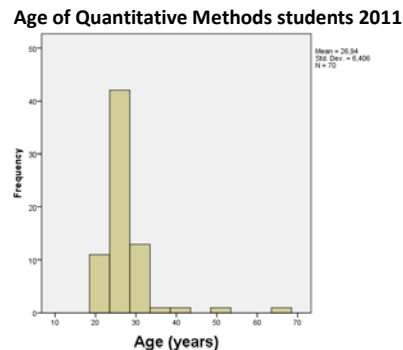


NOTE! Titles cannot be set within the boxplot procedure. Make sure to add informative titles manually after the boxplot is produced.

9.5 Histograms

The distribution of a continuous (numerical/quantitative) variable can be visualized by a histogram.

Histogram



Choose **Graphs >> Legacy Dialogs >> Histograms** from the Menu tab. The dialog window below will appear.

The screenshot shows the SPSS Histogram dialog box. The variable 'Age (years) [Age]' is selected in the 'Variable:' field. The 'Titles...' button is visible. The 'OK' button is highlighted with a circle and an arrow pointing to it from a text box.

1) Add the variable for which you want to produce a histogram for to the Variable field

2) Click "Titles", and type an informative title explaining what the graph is displaying.

3) Click "OK" to produce the graph

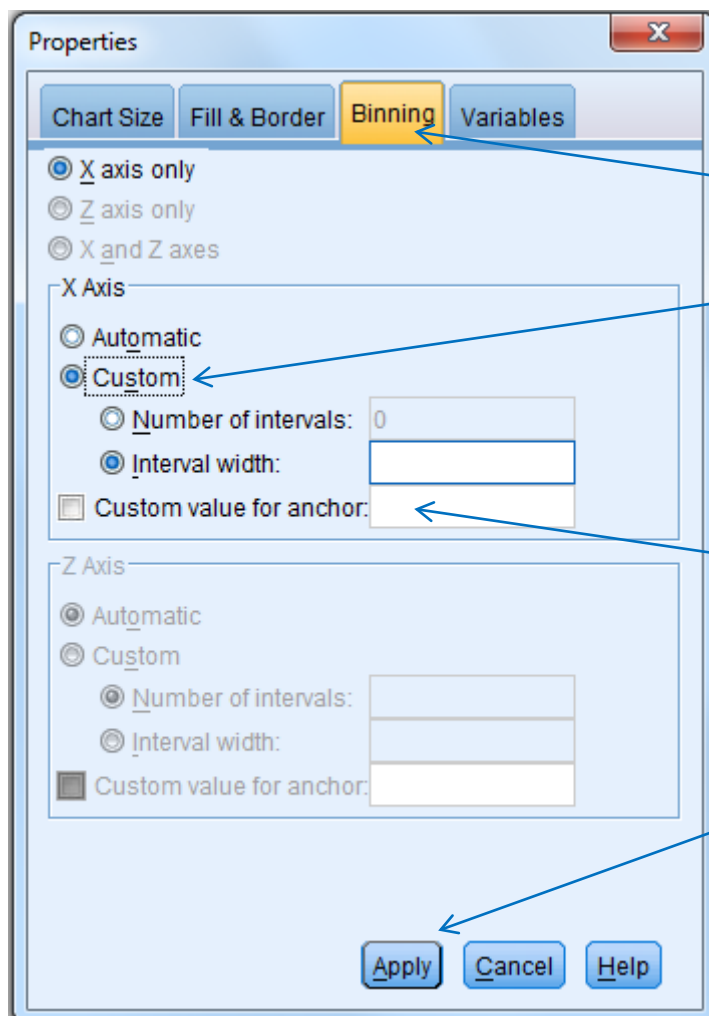
Another way of creating histograms is to use the Chart builder, which is described in **chapter 5 of IBM SPSS Statistics 22 Brief Guide**:

ftp://ftp.software.ibm.com/software/analytics/spss/documentation/statistics/22.0/en/client/Manuals/IBM_SPSS_Statistics_Brief_Guide.pdf

9.5.1 Change the width of the intervals.

You can easily change the width of the intervals.

- 1) Double click the histogram (Output window) to open the Chart Editor (see section 10 for more information on the Chart Editor).
- 2) Double click one of the bars. This will open the Properties window below.



The screenshot shows the 'Properties' dialog box with the 'Binning' tab selected. The 'X axis only' radio button is selected. Under 'X Axis', the 'Custom' radio button is selected, and the 'Interval width' is set to 0. The 'Z Axis' section is also visible, with 'Automatic' selected. The 'Apply' button is highlighted at the bottom.

1) Click the "Binning" tab.

2) Click "Custom" and type either the number of intervals or the interval width that you wish to use.

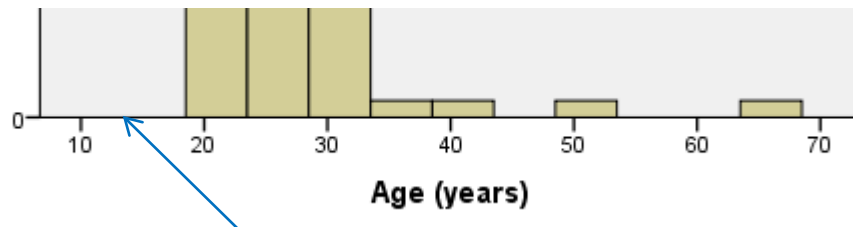
4) If you want the lowest interval to start at a certain value, mark "Custom value for anchor" and type the value.

5) Click "Apply" to make the change(s) effective.

9.5.2 Setting the class limits.

It is important that the class limits in a histogram are clear.

1) Double click the histogram (Output window) to open the Chart Editor (see section 10 for more information on the Chart Editor).



2) Double click the X axis. This will open the Properties window below.

Properties

Labels & Ticks Number Format Variables

Chart Size **Scale** Lines

Range

	Auto	Custom	Data
Minimum	☒	150	150
Maximum	☒	210	201
Major Increment	☐	6	
Origin	☒	0	

☐ Display line at origin

Type

☒ Linear

☐ Logarithmic

Base: 10 ☒ Safe

☐ Power

Exponent: 0,5 ☒ Safe

Lower margin (%): 5 Upper margin (%): 5

Apply Cancel Help

1) Click the "Scale" tab.

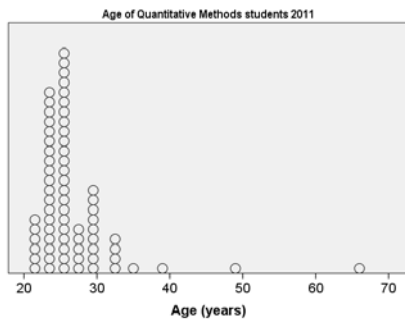
2) Set the "Major Increment" by typing a value that is a multiple of the interval width.

If e.g. the interval width is 3, you can set the major increment to 3, 6, 9, etc.

3) Click "Apply" to make the change(s) effective.

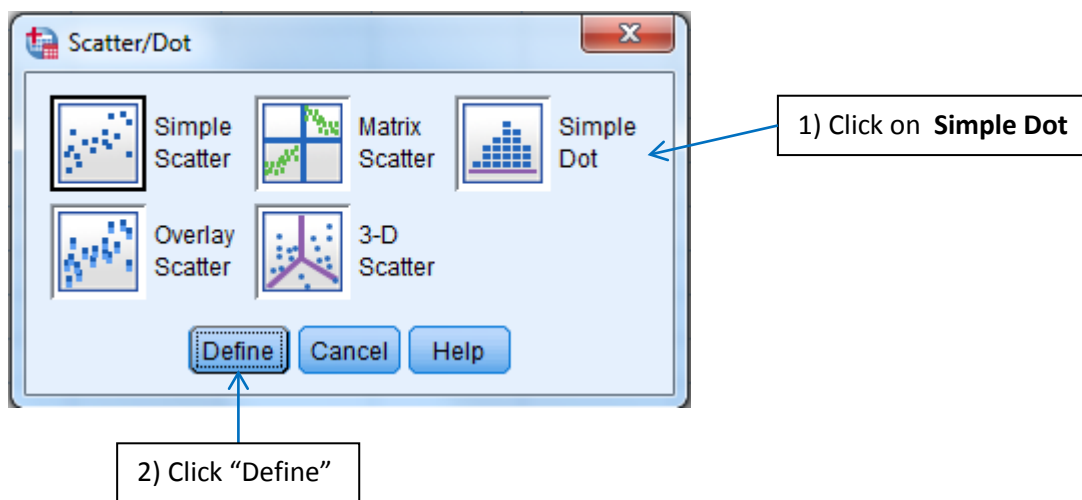
9.6 Dot plots

The distribution of a continuous (numerical/quantitative) or categorical variable can be visualized by a dot plot. In a dot plot, each individual is represented by a dot.

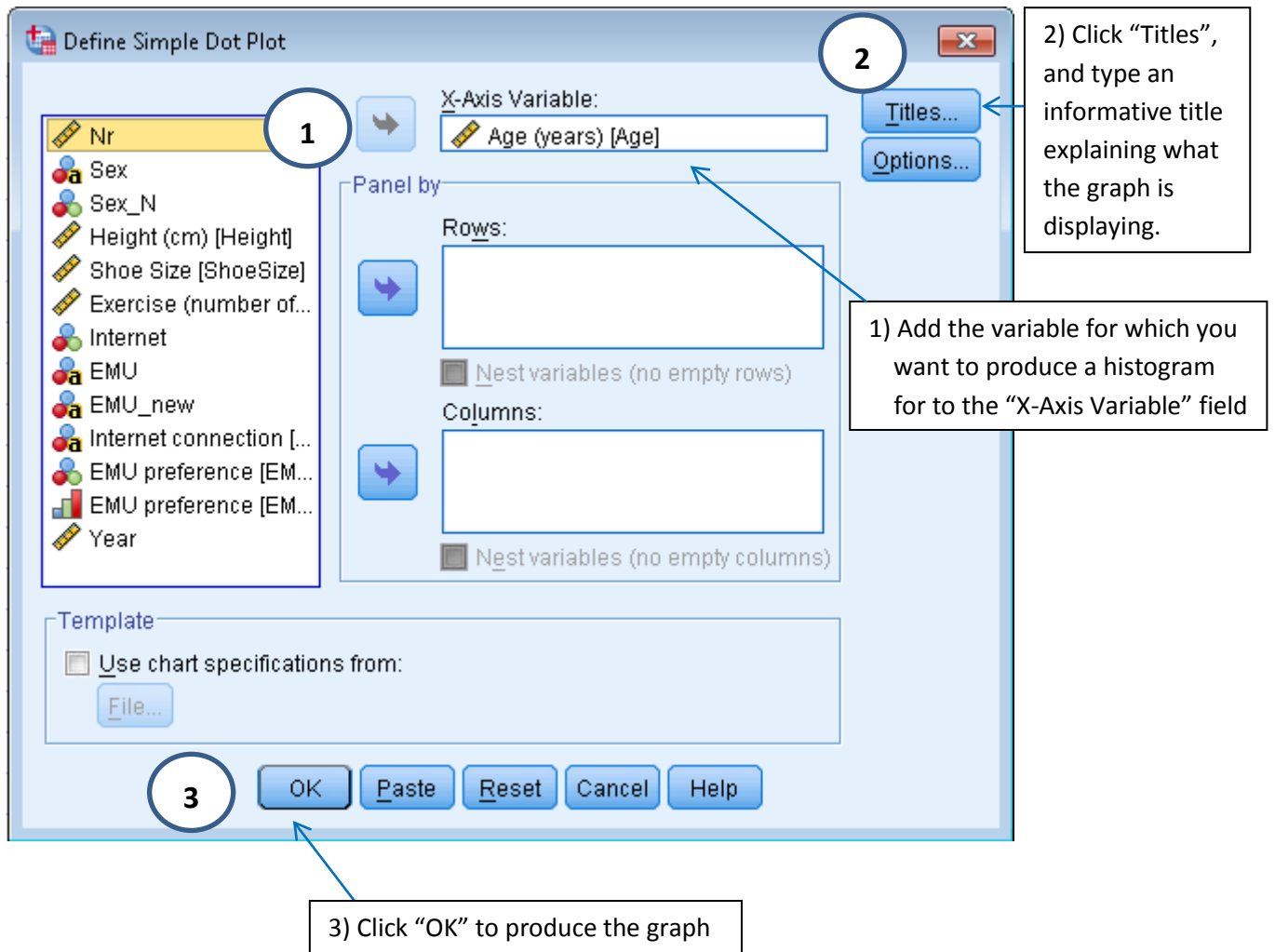


Dot Plot

Choose **Graphs >> Legacy Dialogs >> Scatter/Dot** from the Menu tab. The following dialog window will appear.



The dialog window below will then appear.



You might want to resize the circles, see section 10 for a description of the Chart Editor.

9.7 Stem-and-leaf plots

The distribution of a continuous (numerical/quantitative) variable can be visualized by a stem-and-leaf plot, see below.

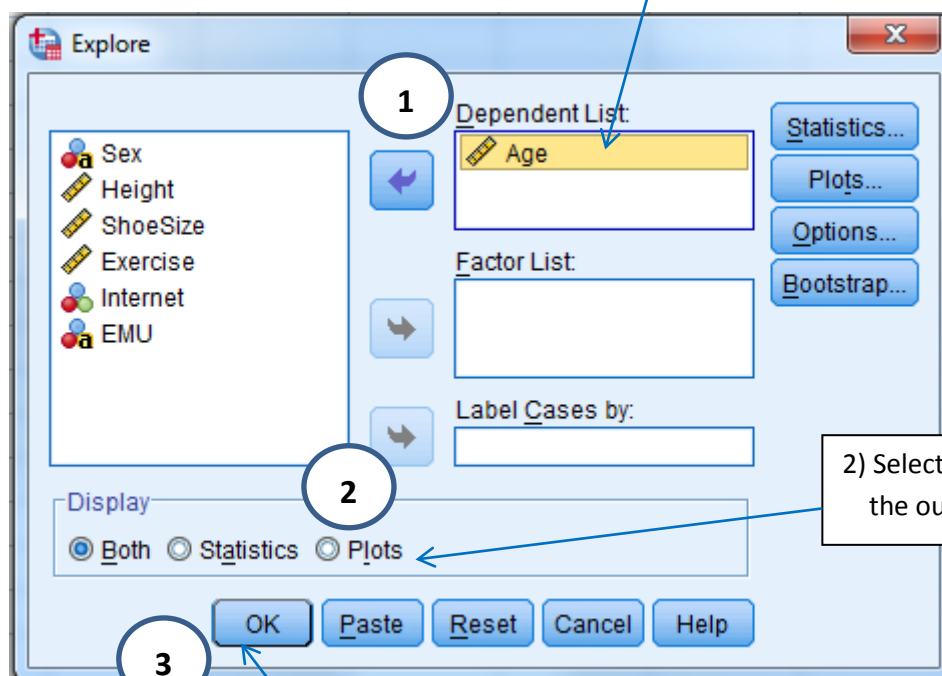
Age (years) Stem-and-Leaf Plot		
Frequency	Stem &	Leaf
2,00	2 .	11
9,00	2 .	222233333
28,00	2 .	444444444444444455555555555555
11,00	2 .	66666666677
9,00	2 .	888999999
3,00	3 .	000
4,00	3 .	2233
4,00	Extremes	(>=35)
Stem width: 10		
Each leaf: 1 case(s)		

The stem width provides information about the size of the values.

In this example the stem width is 10, which means that the first value is 21. If the stem width would have been 1, the first value would be 2.1.

Stem-and-leaf plot

Choose **Analyze >> Descriptive Statistics >> Explore** from the Menu tab. The dialog window below will appear.



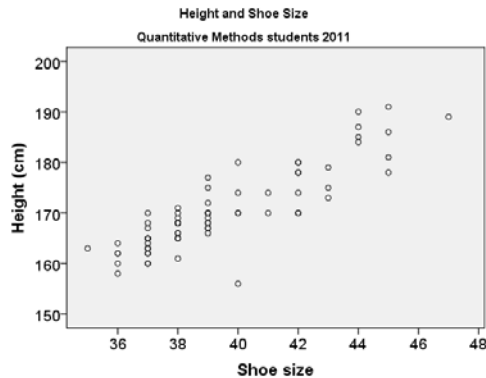
1) Add the variable for which you want to produce a stem-and-leaf-plot to the "Dependent List" field. A stem-and-leaf plot is produced by default.

2) Select "Plots" if you want to reduce the output.

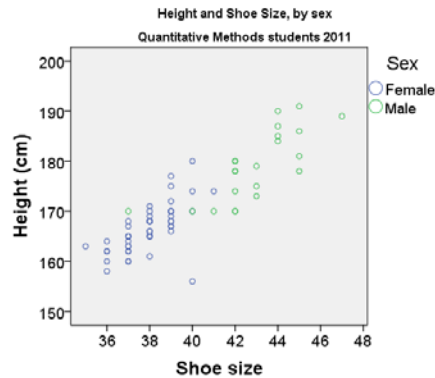
3) Click "OK" to produce the confidence interval

9.8 Scatter plots

To visualize the relationship between two numerical variables, a scatter plot can be produced.



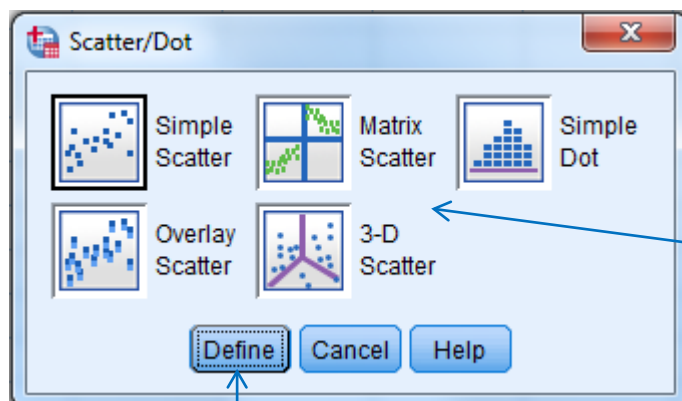
Simple scatter plot



*Simple scatter plot,
markers set by sex*

If one of the variables isn't numerical, or both variables are categorical (but at least one of them can be ordered), the relationship can be visualized by a grouped box plot (see section 9.4 above).

Choose **Graphs >> Legacy Dialogs >> Scatter/Dot** from the Menu tab. The following dialog window will appear.



2) Click "Define"

1) Choose which kind of scatter or dot plot you want to produce.

Simple Scatter: Displays the relationship between two variables.

Overlay Scatter: Displays the relationship between two pairs of variables simultaneously.

Matrix Scatter: Displays several simple scatter plots simultaneously, for different combinations of pairs of variables.

3-D Scatter: Displays the relationship between three variables.

Simple Dot: Displays the distribution of one single variable. Each individual is represented by a circle (dot).

9.8.1 Simple scatter plot

If you choose “Simple” above, the dialog window below will appear.

The screenshot shows the 'Simple Scatterplot' dialog box in SPSS. It features a list of variables on the left: Sex, Height, ShoeSize, Internet, and EMU. The main area contains fields for 'Y Axis:' (set to 'Age'), 'X Axis:' (set to 'Exercise'), 'Set Markers by:', 'Label Cases by:', and 'Panel by' (with 'Rows' and 'Columns' sub-fields). At the bottom, there is a 'Template' section with a checkbox for 'Use chart specifications from:' and a 'File...' button. The 'OK' button is circled with a '3'. Annotations include: '1' pointing to the variable list, '2' pointing to the 'Titles...' button, and '3' pointing to the 'OK' button. Three text boxes provide instructions: '1) Add the variables for which you want to produce a scatter plot.', '2) Click “Titles”, and type an informative title explaining what the graph is displaying.', and '3) Click “OK” to produce the graph'. A note also defines 'Y axis = vertical axis' and 'X axis = horizontal axis'.

1) Add the variables for which you want to produce a scatter plot.

2) Click “Titles”, and type an informative title explaining what the graph is displaying.

Y axis = vertical axis
X axis = horizontal axis

If you want different symbols for different subsets, add the variable you want to base the subsets on.

3) Click “OK” to produce the graph

Scatter plots can also be produced using the Chart builder, described in **chapter 5** of **IBM SPSS Statistics 22 Brief Guide**:

ftp://ftp.software.ibm.com/software/analytics/spss/documentation/statistics/22.0/en/client/Manuals/IBM_SPSS_Statistics_Brief_Guide.pdf

10 Editing graphs with the Chart Editor

You can edit charts in a variety of ways, you can e.g.:

- Change colors
- Edit text
- Display data value labels

Double click on the produced graph to open the Chart Editor.

When you have finished editing, close the Chart Editor to get back to the Output window where the edited graph will be displayed.

10.1 Selecting graph elements

To edit a graph element, you first select it by clicking on any one of the elements of the graph (e.g. on a bar or pie sector). The rectangles around the elements indicate that they are selected.

There are general rules for selecting elements in simple graphs:

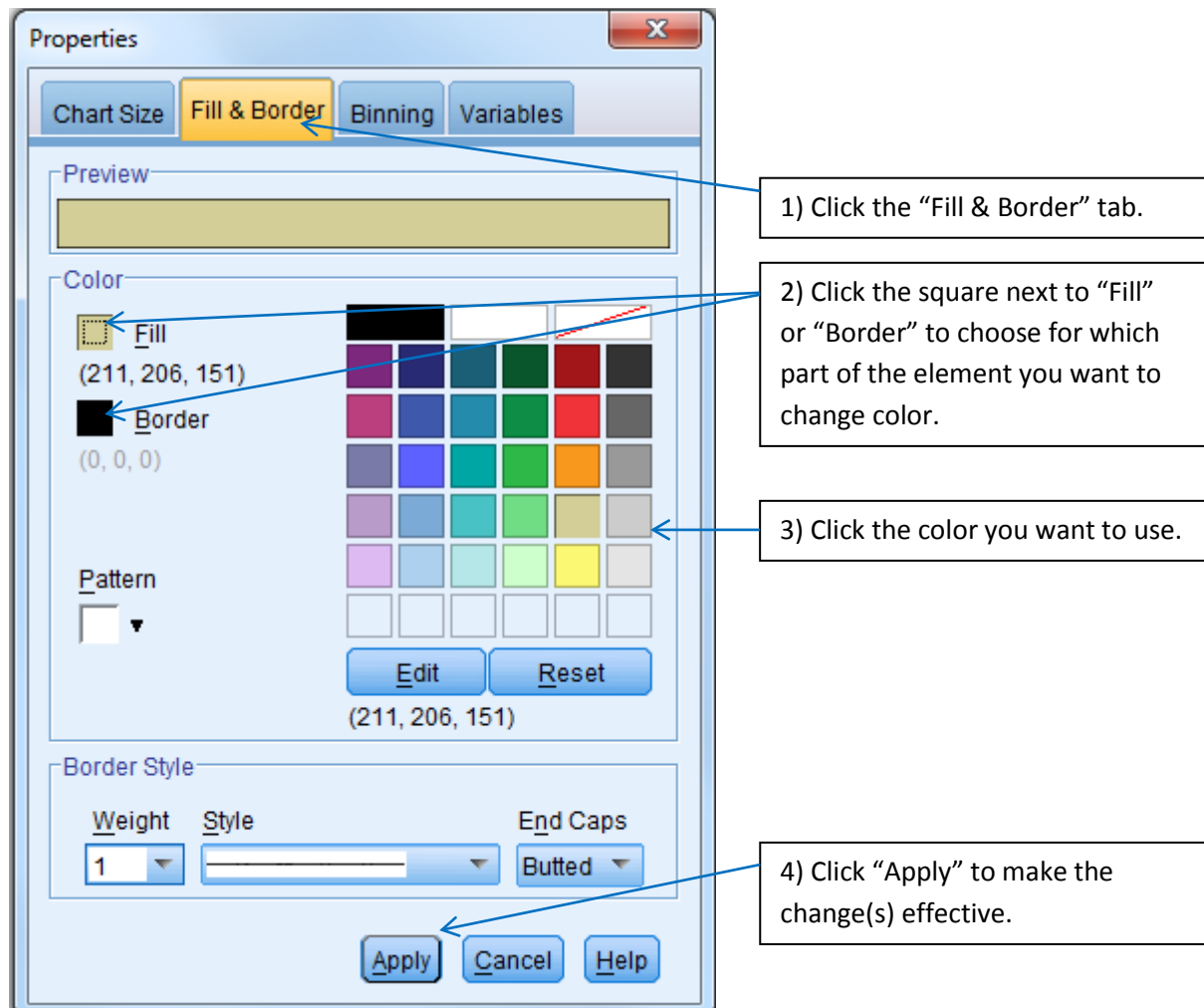
- When no graphic elements are selected, click any graphic element to select all graphic elements.
- When all graphic elements are selected, click a graphic element to select only that graphic element. You can select a different graphic element by clicking it. To select multiple graphic elements, click each element while pressing the Ctrl key.
- To deselect all elements, press the Esc key.
- Click any bar to select all of the bars again.

10.2 Using the Properties window

From the Chart Editor menus choose **Edit > Properties** (You can also use the keyboard shortcut Ctrl+T). This opens the Properties window, showing the tabs that apply to the bars you selected. These tabs change depending on what graph element you select in the Chart Editor. For example, if you had selected a text frame instead of bars, different tabs would appear in the Properties window. You will use these tabs to do most chart editing.

10.3 Changing bar colors

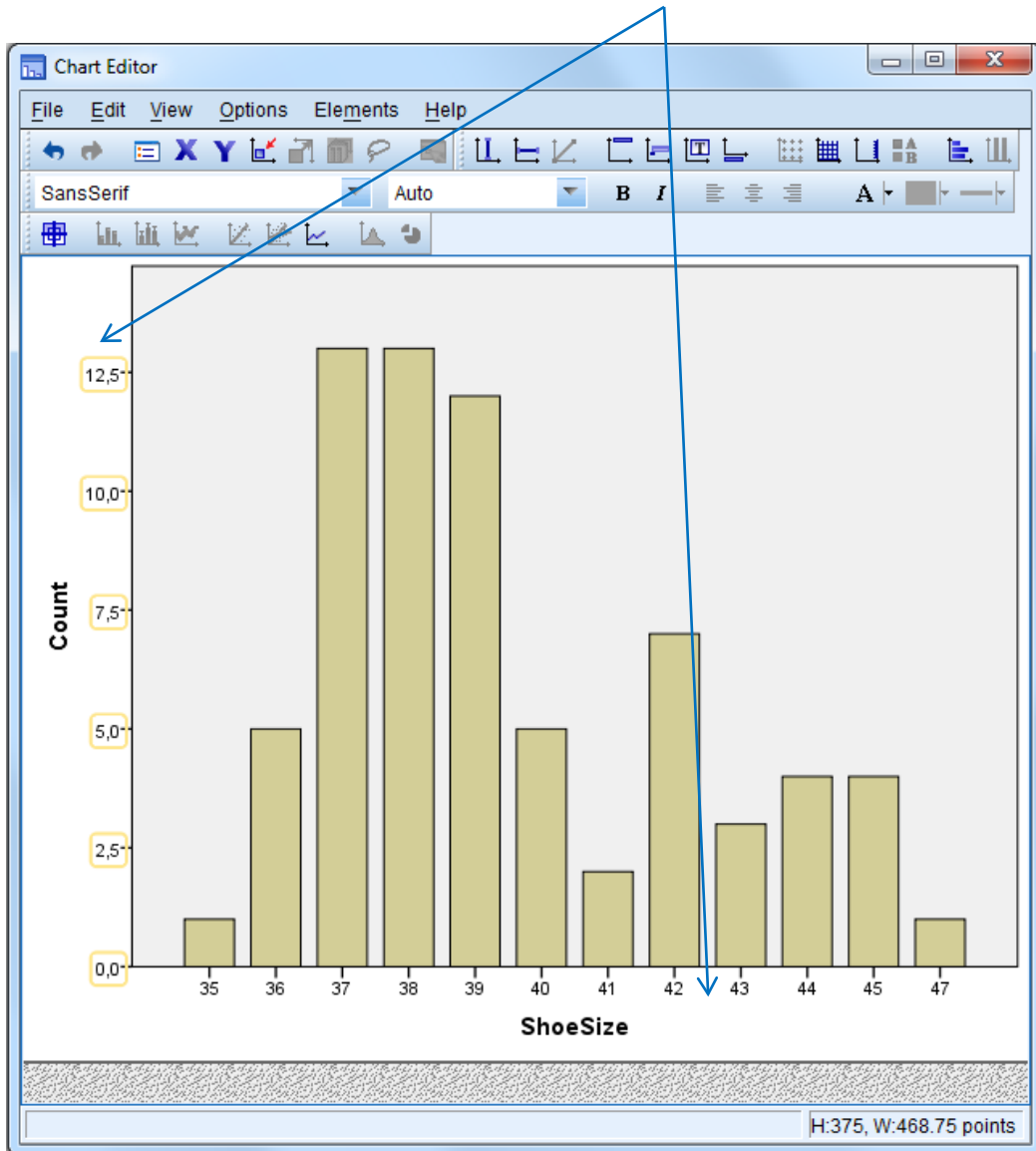
To change the color of the elements in a graph (bars, pie sectors, etc.), you specify color attributes of graphic elements (excluding lines and markers) on the Fill & Border tab in the Properties window (section 10.2 above describes how to find the Properties window). The appearance of the Properties window depends on what kind of graph is being produced. The example below is from the creation of a histogram.



10.4 Formatting numbers in tick labels

If you want to change the scaling of the numbers on the x or y axis, you can change the number format in the tick labels and edit the axis title appropriately.

Select the x or y axis tick labels by clicking any one of them.



Then open the Properties window (see section 10.2 above) and click the “Number Format” tab, see below.

The screenshot shows the 'Properties' dialog box with the 'Number Format' tab selected. The 'Sample' section shows 'The number 1000000 will appear as: 1,000,000'. The 'Decimal Places' is set to 0, 'Scaling Factor' is 1, 'Leading Characters' and 'Trailing Characters' are empty. 'Display Digit Grouping' is checked, and 'Scientific Notation' is set to 'Automatic'. The 'Apply', 'Cancel', and 'Help' buttons are at the bottom.

Click the "Number Format" tab.

If you don't want the tick labels to display decimals, type 0 in the Decimal Places text box.

The Scaling factor is the number by which the Chart Editor divides the displayed number.

E.g., if the numbers on your axis are scaled in hundreds and you want actual numbers, type 0.01 (that will increase the numbers by 100).

"Digit Grouping" means that a comma is used to separate thousands in large numbers.

Unselect "Digit Grouping" if you want the Swedish version without commas.

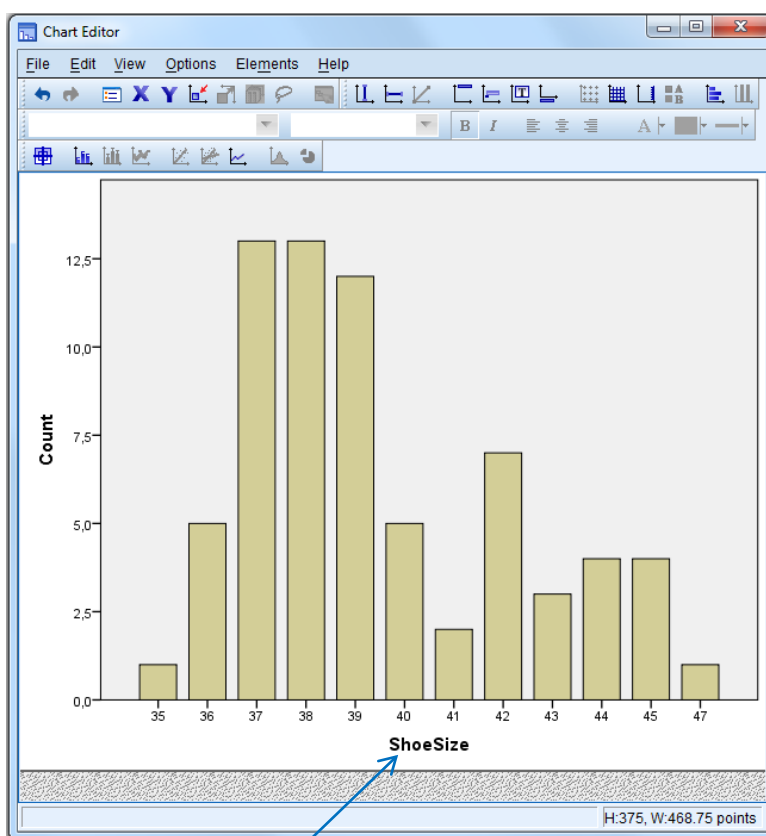
Click "Apply" to make the change(s) effective.

10.5 Editing text

You might want to edit the text in your axis titles for a number of reasons:

- You haven't used value labels (see section 4.2.4) and the variable name is not very informative
- You want to add units (cm, kg, etc.)
- If you change the number format of the tick labels (see section 10.4 above), the axis title may no longer be accurate and you have to change it to reflect the new number format

Note: You do not need to open the Properties window to edit text. You can edit text directly on the chart.



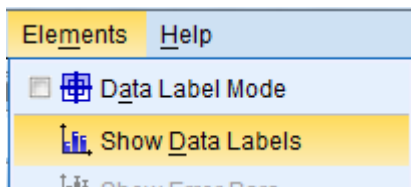
- 1) Click the axis title to select it.
- 2) Click the axis title again to start edit mode. While in edit mode, the Chart Editor positions any rotated text horizontally. It also displays a flashing red bar cursor (not shown in the example).
- 3) Edit the text as you wish (delete existing text, add new text)
- 4) Press Enter to exit edit mode and update the axis title.

10.6 Displaying data value labels

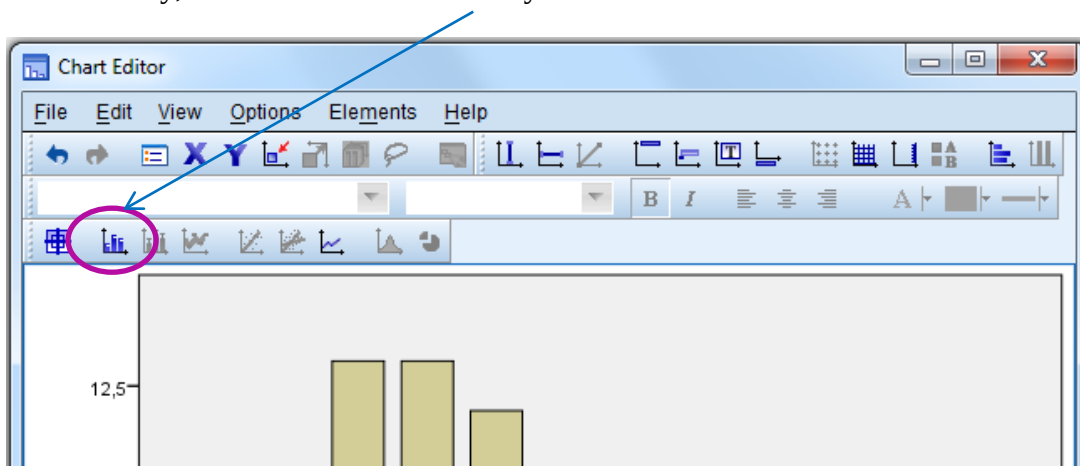
You can, if you wish, show the exact values associated with the graphic elements (bars, pie sectors, etc.). These values are displayed in data labels.

Double click on the graph to open the Chart Editor.

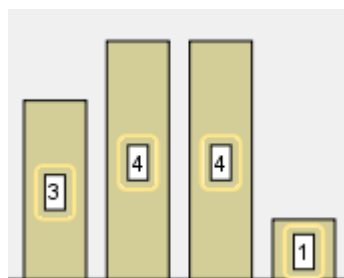
Choose **Elements > Show Data Labels** from the Menu bar.



Alternatively, click on the Data Labels symbol



The exact values of each element in the graph are now displayed.



11 Descriptive statistics

11.1 Simple descriptive statistics and frequency tables

To produce simple descriptive statistics, follow the instructions in chapter 4 of **IBM SPSS**

Statistics 22 Brief Guide:

ftp://ftp.software.ibm.com/software/analytics/spss/documentation/statistics/22.0/en/client/Manuals/IBM_SPSS_Statistics_Brief_Guide.pdf

With these instructions you'll learn how to:

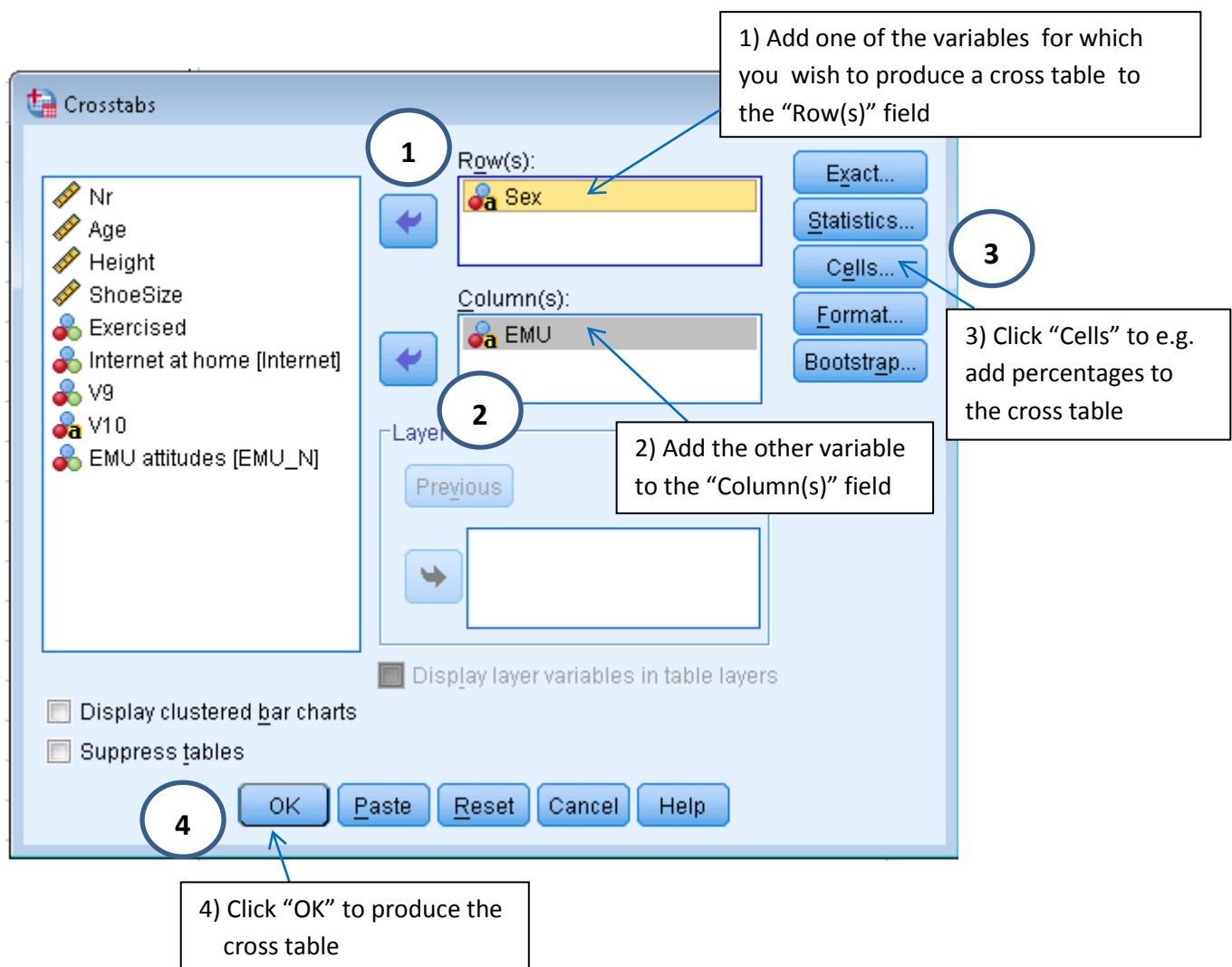
- Produce summary measures for numerical/scale variables
- Produce summary measures for categorical data

11.2 Present descriptive statistics for separate groups (split file)

To present statistics for separate groups, split your data file into separate groups for analysis by choosing **Data >> Split File** from the Menu bar (as described in section 5.10.2).

11.3 Two-way frequency tables (cross tables)

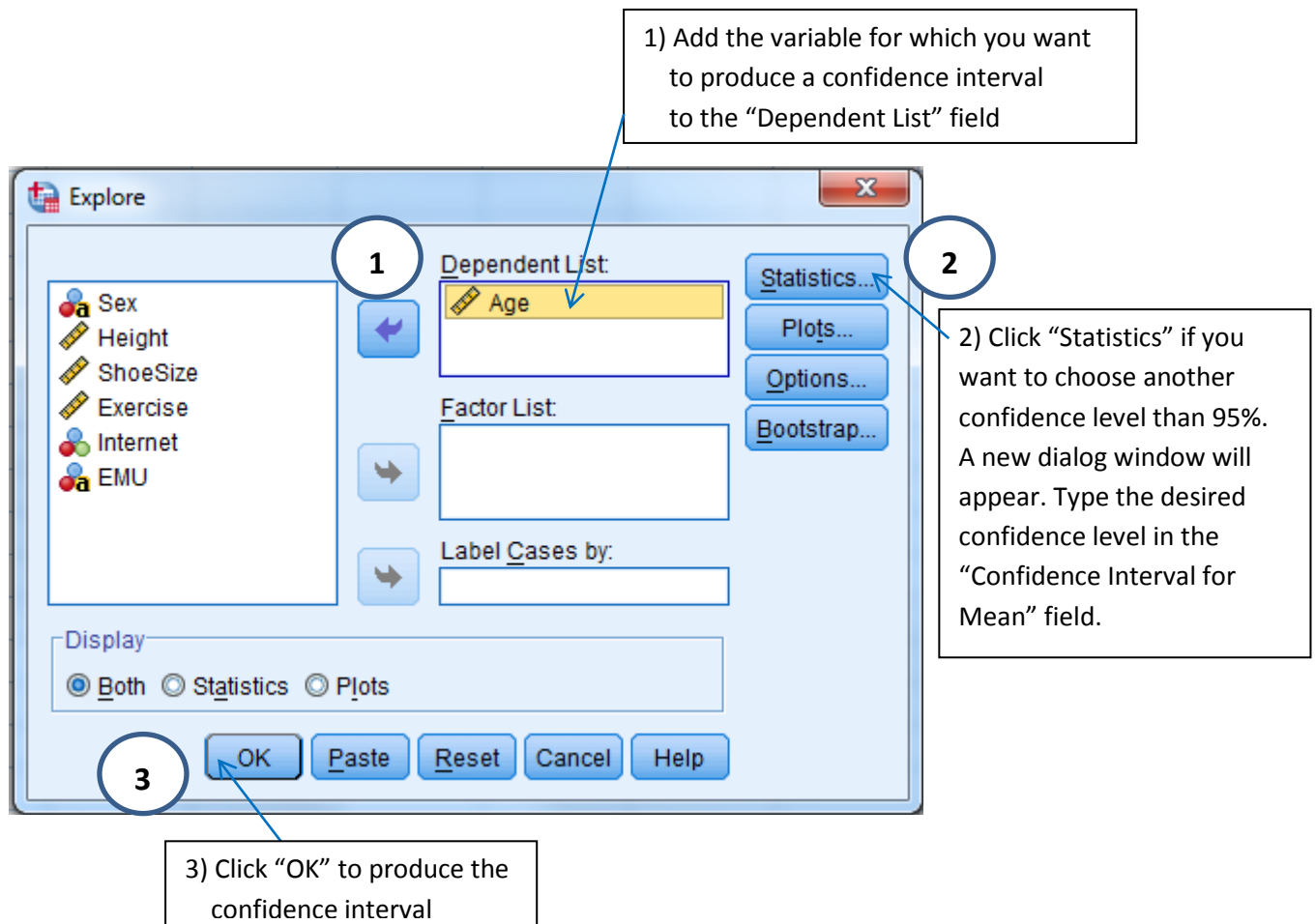
To produce a two-way frequency table (cross table), choose **Analyze >> Descriptive Statistics >> Crosstabs** from the Menu bar. The following dialog window will appear.



12 Confidence intervals

12.1 Confidence interval around a mean

To create a confidence interval around a mean, choose **Analyze >> Descriptive Statistics >> Explore** from the Menu bar. The following dialog window will appear.



The output below should be given:

Case Processing Summary

	Cases					
	Valid		Missing		Total	
	N	Percent	N	Percent	N	Percent
Age	70	100,0%	0	0,0%	70	100,0%

The first table tells the sample size and whether any of your data have been omitted (due to missing values)

Descriptives

		Statistic	Std. Error
Age	Mean	2975,96	2835,775
	95% Confidence Interval for Mean	-2681,26	
	Lower Bound		
	Upper Bound	8633,17	
	5% Trimmed Mean	74,08	
	Median	26,00	
	Variance	562913486,216	
	Std. Deviation	23725,798	
	Minimum	21	
	Maximum	198608	
	Range	198587	
	Interquartile Range	5	
	Skewness	8,362	,287
	Kurtosis	69,946	,566

The upper and lower limits ("bounds") of the confidence interval are presented for the chosen variable.

The confidence intervals are based on the *t*-distribution.

12.2 Confidence interval around a proportion

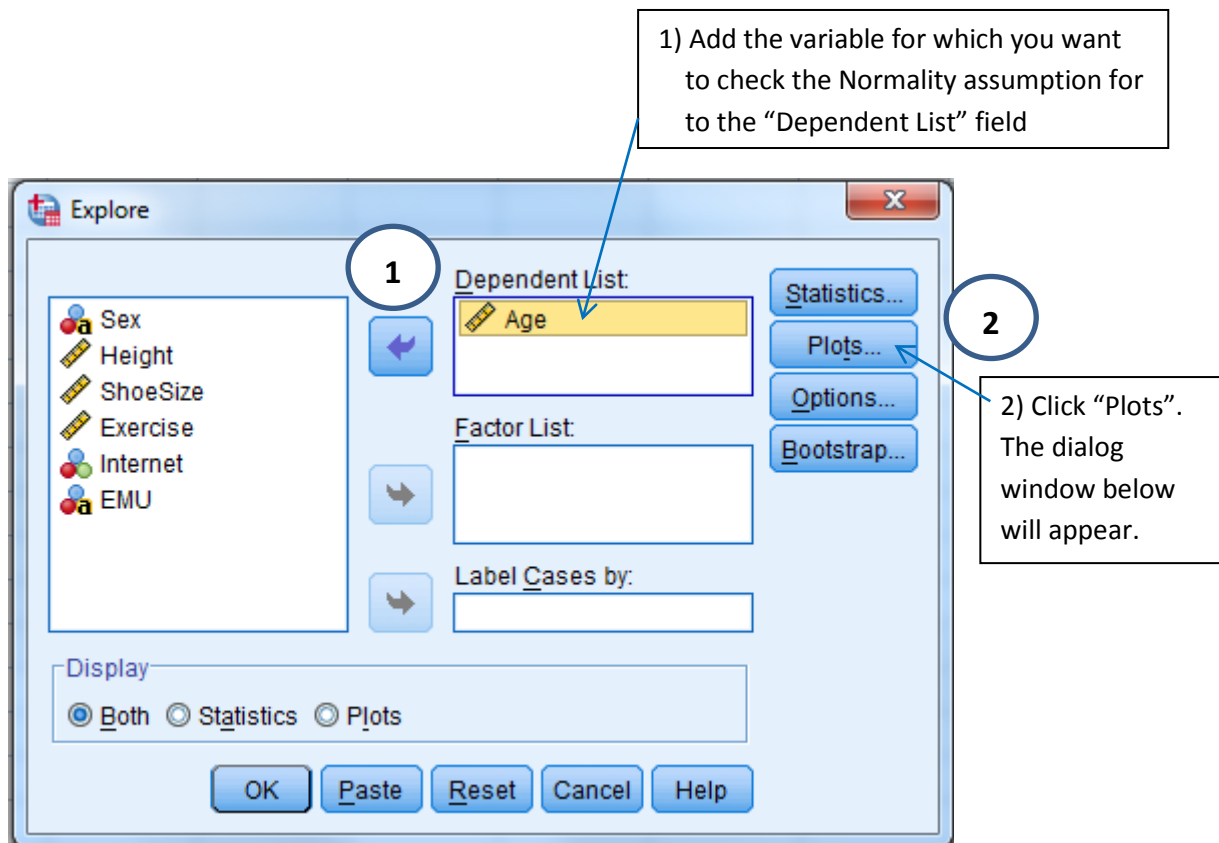
SPSS cannot calculate confidence intervals around proportions; this will have to be done manually.

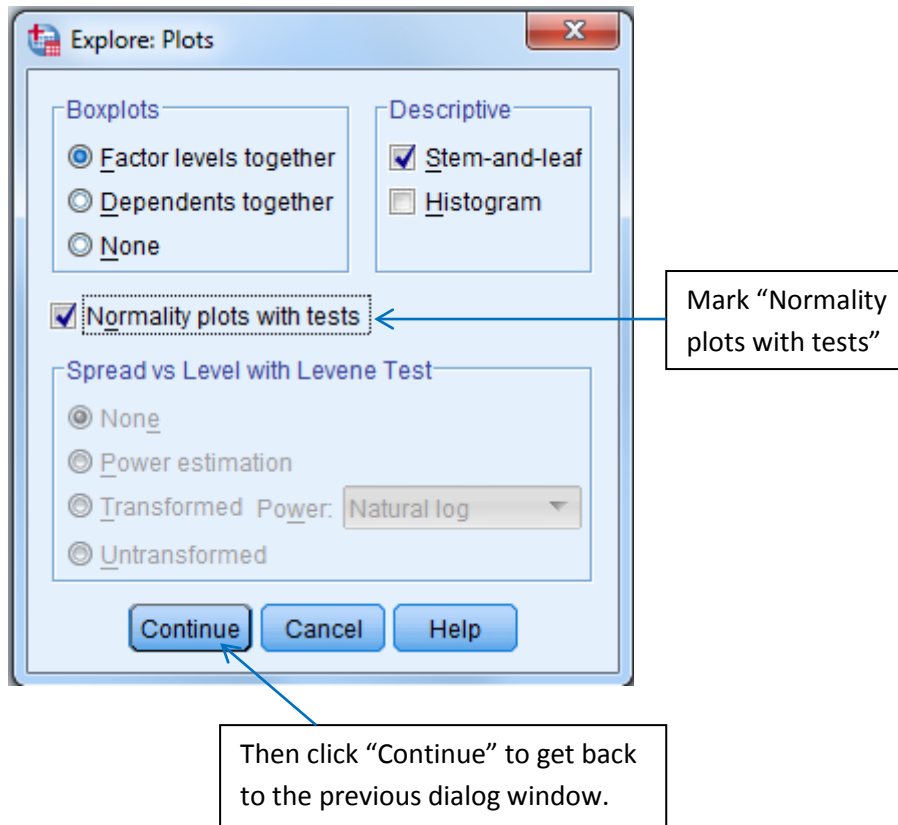
To be able to use SPSS to calculate proportions, make sure you have a variable that only can take values 1 and 0 (where 1 represents the property of interest)

For a variable that only can take values 1 and 0, the mean of that variable represents the proportion of observations with value 1.

13 Normality plots and tests

To check if your variable can be assumed to follow a Normal distribution, you can produce Normality plots and tests by choosing **Analyze >> Descriptive Statistics >> Explore** from the Menu bar. The following dialog window will appear.





Then click "OK" to produce the Normality plots and tests. The following output should be given (in addition to the output presented in section 12.1 above):

Tests of Normality						
	Kolmogorov-Smirnov ^a			Shapiro-Wilk		
	Statistic	df	Sig.	Statistic	df	Sig.
Age	,502	70	,000	,104	70	,000

a. Lilliefors Significance Correction

P-values of the Kolmogorov-Smirnov test and the Shapiro-Wilk test (both are tests of Normality). If **any** of the two *P*-values is <0.05, the null hypothesis of Normality is rejected.

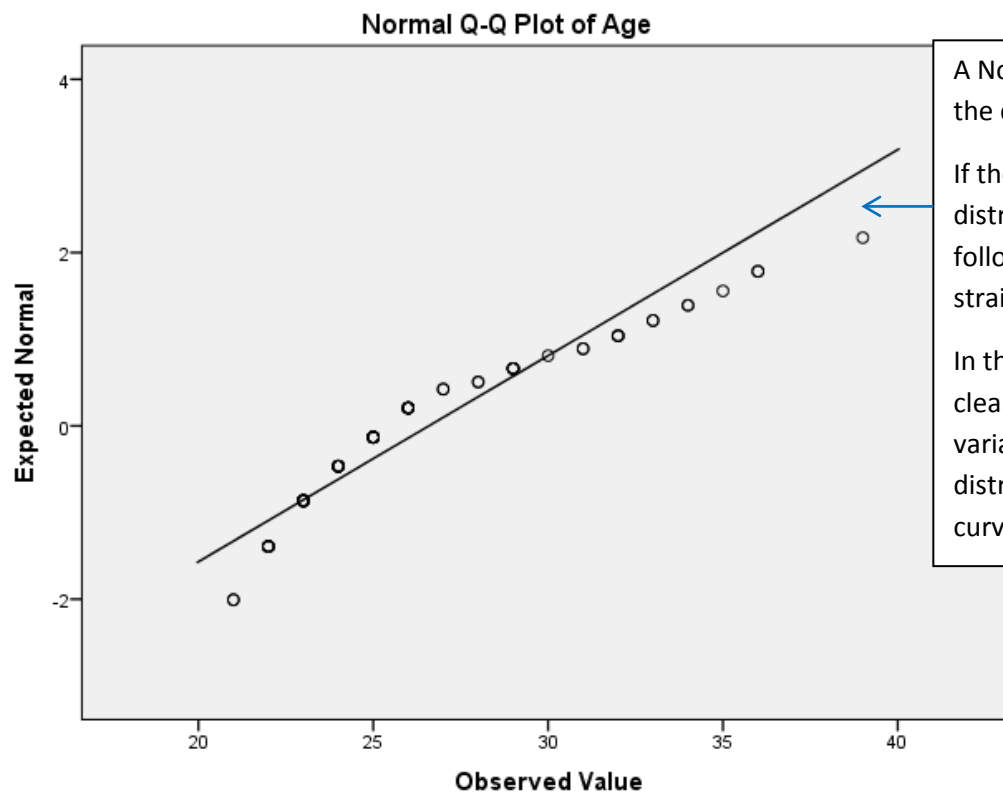
Age Stem-and-Leaf Plot

Frequency	Stem &	Leaf
2,00	2 .	11
15,00	2 .	222222333333333
17,00	2 .	4444444455555555
11,00	2 .	66666666677
7,00	2 .	8899999
3,00	3 .	011
5,00	3 .	22233
3,00	3 .	445
2,00	3 .	66
1,00	Extremes	(>=39)

Stem width: 10
Each leaf: 1 case(s)

A stem-and-leaf plot of the chosen variable. Shows if the variable values are symmetrically distributed.

In this example you can clearly see that the distribution is not symmetrical (might be easier to see if you lean your head to the right).



A Normal quantile plot of the chosen variable.

If the variable is Normally distributed, the circles follow a reasonably straight line.

In this example you can clearly see that the variable is not Normally distributed, there is a curvilinear pattern.

14 One, two and paired samples t-test

To perform t - tests, follow the instructions in **chapter 9** of **IBM SPSS Statistics Base 22**:
ftp://ftp.software.ibm.com/software/analytics/spss/documentation/statistics/22.0/en/client/Manuals/IBM_SPSS_Statistics_Base.pdf

With these instructions you'll learn how to:

- Perform a one-sample t -test and produce a confidence interval around the mean
- Perform a two-sample t -test
- Perform a paired-samples t -test

15 Test of one proportion

To perform a test of one proportion, there are three different approaches:

- 1) Z-test (large samples), which has to be calculated manually since there is no option for z-test of a proportion in SPSS.
- 2) Chi-square test (two-sided hypotheses and large samples), described in section 15.1 below.
- 3) Binomial test (one-sided hypothesis and small or large samples), described in section 15.2 below.

15.1 Two-sided hypotheses and large samples (chi-square test).

To perform a test of one proportion, with two-sided hypotheses and large samples, choose **Analyze >> Non-Parametric tests >> Legacy Dialogs >> Chi-square** from the Menu bar. The dialog window below will appear.

1) Add the variable for which you want to perform a Chi-square test to the "Test Variable List" field.

2) Click "Exact" and select "Exact" in the dialog window that will appear (and then click "Continue").

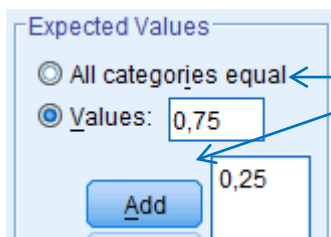
3) Select "Values" and type your null hypothesis proportion (see below).

NOTE! If you want to test that the proportion differs from 50%, let "All categories equal" be selected.

In this example we want to test the hypothesis $H_0: p = 0.75$ against a two-sided alternative.

First type the value of $(1-p_0)$, where (p_0) is the null hypothesis proportion. Click "Add". The value you typed will then be added to the white field.

Then you also have to add the value of p_0 .



Type the value of p_0 (your null hypothesis proportion) and click “Add”.
The value you typed will then be added to the white field.

Then click “OK” to perform the test. The following output should be given:

Sex_N			
	Observed N	Expected N	Residual
0	22	17,5	4,5
1	48	52,5	-4,5
Total	70		

Observed number of individuals in each of the two categories (in this example 0=Male, 1=Female)

Number of individuals expected in each of the two categories according to the null hypothesis.

IMPORTANT! Make sure to check that you entered the proportions correctly by calculating the expected frequencies to see that they are consistent with what is presented in the output.

In this example, Females are denoted by 1. If you want to test e.g. $H_0 : p=0.75$ (where p =the proportion in the category denoted by 1, i.e. the proportion of Females), then the expected frequency of Females should be $40 \times 0,75 = 52.5$ according to the null hypothesis.

Test Statistics	
	Sex_N
Chi-Square	1,543 ^a
df	1
Asymp. Sig.	,214
Exact Sig.	,269
Point Probability	,103

a. 0 cells (0,0%) have expected frequencies less than 5. The minimum expected cell frequency is 17,5.

The value of the test statistic, “Chi-square”.

df = number of categories -1

Asymptotic P -value (based on large-sample properties). Should only be used if the exact P -value cannot be calculated by SPSS.

Exact P -value. Always use this value if possible (it will not be included in the table if it cannot be calculated).

If any of the “cells” (i.e. categories in this case) has an expected frequency smaller than 5

(which is an assumption for this test to be valid) it will be noted in the footnote above.

15.2 One-sided hypotheses (Binomial test).

To perform a test of one proportion, with two-sided hypotheses and samples of any size, follow the instructions in **IBM SPSS Statistics Base 22 pp. 129-**

130: ftp://ftp.software.ibm.com/software/analytics/spss/documentation/statistics/22.0/en/client/Manuals/IBM_SPSS_Statistics_Base.pdf

16 Test of three or more proportions for a single variable (frequency table)

If you have a variable with three or more categories, you can test if the proportion of individuals/cases with a certain characteristic differs between the categories (or if the proportion is the same in all categories).

Example: A randomly chosen package of 100 M&Ms contained the following candies:

12 red

16 blue

15 yellow

14 orange

20 green

23 brown

To test the null hypothesis that the proportions of red, blue, yellow, orange, green and brown candies are the same (evenly distributed), a chi-square test can be used.

NOTE! The variable has to be numerical to perform a chi-square test. In this example a variable named Color_Numerical has been created, with variable values 1 to 6. Value labels have then been defined as described in section 4.2.4.

Choose **Analyze >> Non-Parametric tests >> Legacy Dialogs >> Chi-square** from the Menu bar. Follow the instructions described in section 15.1 above. In step 3; select “All categories equal”. The output below should then be given:

Color_Numerical

	Observed N	Expected N	Residual
Red	12	16,7	-4,7
Blue	16	16,7	-,7
Yellow	15	16,7	-1,7
Orange	14	16,7	-2,7
Green	20	16,7	3,3
Brown	23	16,7	6,3
Total	100		

Observed number of cases in each of the different categories.

Number of cases expected in each of the categories according to the null hypothesis (even distribution).

Test Statistics

	Counts
Chi-Square	5,000 ^a
df	5
Asymp. Sig.	,416
Exact Sig.	,425
Point Probability	,015

The value of the test statistic, "Chi-square".

df = number of categories - 1

Asymptotic *P*-value (based on large-sample properties). Should only be used if the exact *P*-value cannot be calculated by SPSS.

Exact *P*-value. Always use this value if possible (it will not be included in the table if it cannot be calculated).

a. 0 cells (0,0%) have expected frequencies less than 5. The minimum expected cell frequency is 16,7.

17 Tests of two groups' proportions, chi-squared tests (two-way tables)

The Chi-Square test is a statistical tool used to examine the relationships between nominal or categorical variables.

To produce two-way tables choose **Analyze >> Descriptive Statistics >> Crosstabs** from the menu bar. The following window will appear.

The screenshot shows the SPSS Crosstabs dialog box. The 'Row(s):' field contains 'Gender [Gender]' and the 'Column(s):' field contains 'Statistics Course [B1]'. The 'Statistics...' button is highlighted. The 'Exact...' button is also visible. The 'Cells...' button is highlighted. The 'Format...' button is visible. The 'Style...' button is visible. The 'Display clustered bar charts' checkbox is checked. The 'Suppress tables' checkbox is checked. The 'OK' button is highlighted.

1) Add the first variable to be analyzed to the "Rows" field.

2) Insert the second variable to be analyzed to the "Columns" field.

3) Click the "Exact" button, and tick the "Exact" box in the dialog window that will appear. This way you can request Fisher's exact test for small samples.

4) Click "Statistics" and select "Chi-Square" in the dialog window that will appear. If the variables are nominal, under the "Nominal" column choose "Phi and Cramer's V".

5) Click "Cells" to add percentages to the crosstabulation. Tick the boxes for the type of percentages you wish: row, column and/or total.

6) Click "OK" to produce the test result

The **Case Processing Summary** presented in the output window tells us what proportion of the observations had non-missing values for both Gender and Statistics Course.

The second obtained table **Gender *Statistics Course Crosstabulation** contains the crosstabulation (see below). We can quickly observe information about the interaction of these two variables. If the row variable is Gender and the column variable is Statistics Course, then the row percentage will tell us what percentage of the males or what percentage of the females chose a different course. That is, variable Gender will determine the denominator of

the percentage computations.

Gender * Statistics Course Crosstabulation

			Statistics Course			Total
			Grundläggande statistik A4	Statistik för ekonomer A8	Statistik för samhällsplanerare	
Gender	Female	Count	49	34	13	96
		% within Gender	51,0%	35,4%	13,5%	100,0%
		% within Statistics Course	49,0%	45,9%	56,5%	48,7%
		% of Total	24,9%	17,3%	6,6%	48,7%
	Male	Count	51	40	10	101
		% within Gender	50,5%	39,6%	9,9%	100,0%
		% within Statistics Course	51,0%	54,1%	43,5%	51,3%
		% of Total	25,9%	20,3%	5,1%	51,3%
Total	Count	100	74	23	197	
	% within Gender	50,8%	37,6%	11,7%	100,0%	
	% within Statistics Course	100,0%	100,0%	100,0%	100,0%	
	% of Total	50,8%	37,6%	11,7%	100,0%	

In the third obtained table **Chi-Square Tests** (see below), you will mainly look at the Pearson Chi-Square. When the P-value (presented as “Exact sig” in the table) is less than the significance level, there is a significant relationship between the variables. The presented table below shows that there is no significant relation between Gender and the tendency of choosing a specific Statistics Course, since the P-value of 0.696 is larger than the significance level which means that the null hypothesis of no relationship is not rejected.

Chi-Square Tests

	Value	df	Asymp. Sig. (2-sided)	Exact Sig. (2-sided)	Exact Sig. (1-sided)	Point Probability
Pearson Chi-Square	,791 ^a	2	,673	,696		
Likelihood Ratio	,793	2	,673	,696		
Fisher's Exact Test	,802			,696		
Linear-by-Linear Association	,099 ^b	1	,753	,758	,416	,078
N of Valid Cases	197					

a. 0 cells (0,0%) have expected count less than 5. The minimum expected count is 11,21.

b. The standardized statistic is -,315.

In the fourth table you will get the **Symmetric Measures** (see below). You would normally use Phi for 2X2 tables and Cramer's V for larger tables. Both range from 0 to 1 with 0

representing no relationship between the variables. Here the P-value is > 0.05 ("Approx. Sig."=0.673) which means that the results are not interpretable.

Symmetric Measures

		Value	Approx. Sig.
Nominal by Nominal	Phi	,063	,673
	Cramer's V	,063	,673
N of Valid Cases		197	

18 Non-parametric tests

18.1 Two-sample Wilcoxon Rank Sum test

To perform two-sample Wilcoxon Rank Sum tests (to test the difference between two groups' medians, or systematic differences), choose **Analyze >> Non-Parametric tests >> Legacy Dialogs >> 2 Independent Samples** from the Menu bar. The dialog window below will appear.

The screenshot shows the 'Two-Independent-Samples Tests' dialog box in SPSS. The 'Test Variable List' contains 'Age'. The 'Grouping Variable' is 'Female(1 0)'. The 'Test Type' section has 'Mann-Whitney U' selected. The 'OK' button is highlighted. Numbered annotations (1-5) provide step-by-step instructions for configuring the test.

- 1) Add the variable for which you want to perform a Wilcoxon Rank Sum test to the "Test Variable List" field.
- 2) Select the variable that defines the groups to be compared. Click "Define Groups" and type the values that denote the two groups in the dialog window that will appear. NOTE! The grouping variable has to be numerical.
- 3) Click "Exact" and select "Exact" in the dialog window that will appear (and then click "Continue").
- 4) Ensure that "Mann-Whitney U" is selected.
- 5) Click "OK" to produce the test result

18.2 Wilcoxon Signed Ranks test for paired/matched observations

To perform two-sample Wilcoxon Rank Sum tests (to test the difference between two groups' medians, or systematic differences), choose **Analyze >> Non-Parametric tests >> Legacy Dialogs >> 2 Related Samples** from the Menu bar. The dialog window below will appear.

The screenshot shows the 'Two-Related-Samples Tests' dialog box in SPSS. The 'Test Pairs:' table contains two rows: Pair 1 with Variable1 '[Round1]' and Variable2 '[Round2]', and Pair 2 which is empty. The 'Test Type' section has the 'Wilcoxon' checkbox selected. The 'Exact...' button is highlighted. The 'OK' button is at the bottom left. Numbered annotations (1-4) provide step-by-step instructions for configuring the test.

1) Add the two variables that contain the paired observations to the "Test Pairs" field.

2) Click "Exact" and select "Exact" in the dialog window that will appear (and then click "Continue").

3) Ensure that "Wilcoxon" is selected.

4) Click "OK" to produce the test result

Pair	Variable1	Variable2
1	[Round1]	[Round2]
2		

Test Type

- ☒ Wilcoxon
- ☐ Sign
- ☐ McNemar
- ☐ Marginal Homogeneity

18.3 Kruskal-Wallis test for 3 or more independent groups

To perform K-sample Kruskal-Wallis tests (to test the difference between 3 or more groups' medians, or systematic differences), choose **Analyze >> Non-Parametric tests >> Legacy Dialogs >> K Independent Samples** from the Menu bar. The dialog window below will appear.

The screenshot shows the 'Tests for Several Independent Samples' dialog box in SPSS. The 'Test Variable List' contains 'Height (cm) [Height]'. The 'Grouping Variable' is 'Group(1 3)' with the 'Define Range...' button next to it. Under 'Test Type', 'Kruskal-Wallis H' is selected. The 'OK' button is at the bottom left.

1) Add the variable for which you want to perform a Kruskal Wallis test to the "Test Variable List" field.

2) Select the variable that defines the groups to be compared. Click "Define Range" and type the range of values that denote the different groups, in the dialog window that will appear. NOTE! The grouping variable has to be numerical.

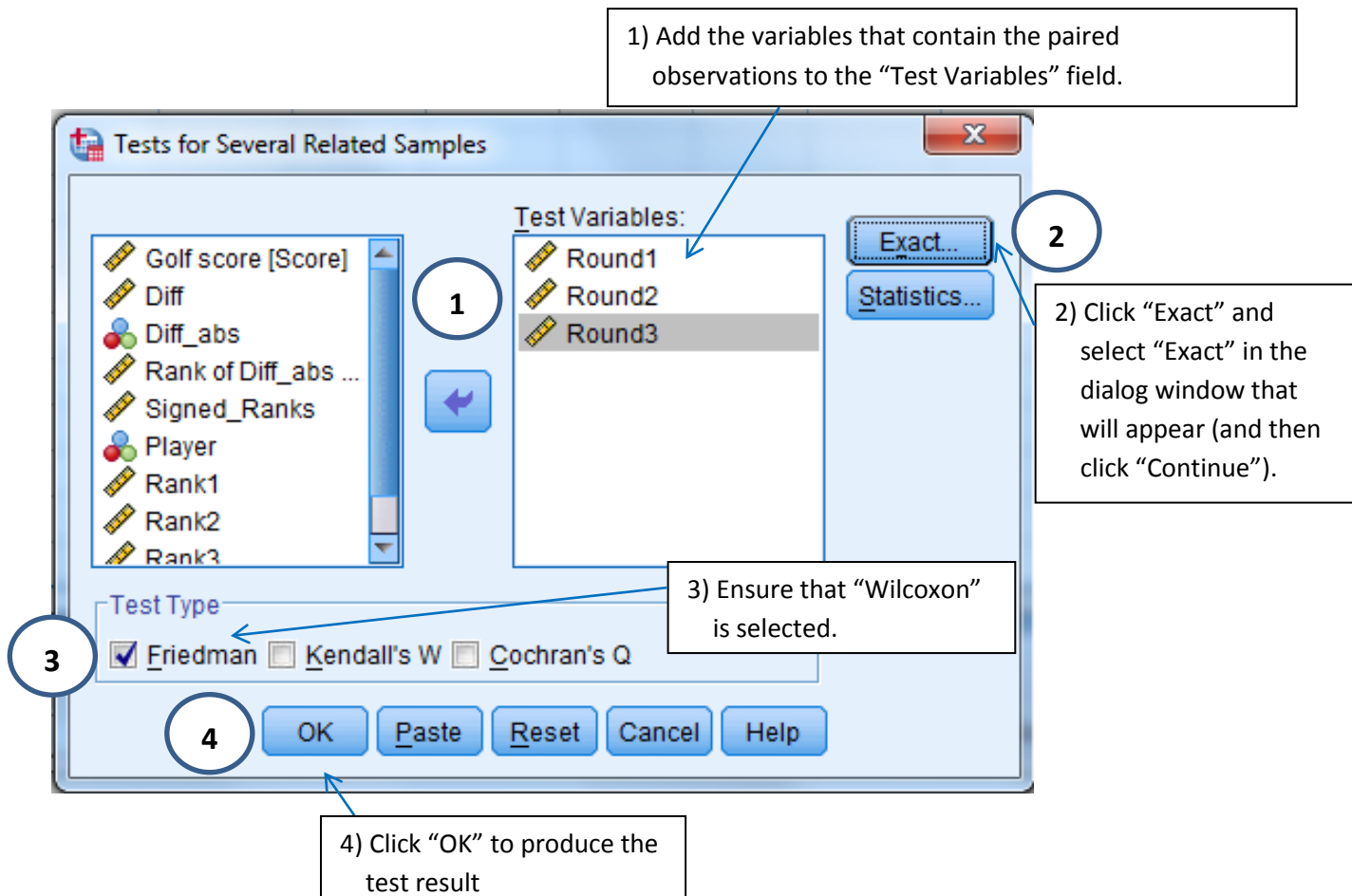
3) Click "Exact" and select "Exact" in the dialog window that will appear (and then click "Continue").

4) Ensure that "Kruskal-Wallis H" is selected.

5) Click "OK" to produce the test result

18.4 Friedman's test for paired/matched observations

To perform K-sample Friedman's tests (to test the difference between 3 or more groups' medians, or systematic differences), choose **Analyze >> Non-Parametric tests >> Legacy Dialogs >> K Related Samples** from the Menu bar. The dialog window below will appear.



19 Correlation and simple linear regression

19.1 Correlation coefficients

To calculate correlation coefficients, choose **Analyze >> Correlate >> Bivariate** from the Menu bar. The following dialog window will appear.

The screenshot shows the 'Bivariate Correlations' dialog box in SPSS. It includes a list of variables on the left, a 'Variables:' field on the right, and sections for 'Correlation Coefficients', 'Test of Significance', and 'Flag significant correlations'. Numbered annotations (1-5) point to specific elements: 1) The 'Variables:' field; 2) The 'Pearson' checkbox under 'Correlation Coefficients'; 3) The 'Two-tailed' radio button under 'Test of Significance'; 4) The 'Options...' button; 5) The 'OK' button.

1) Add the variables for which you want to calculate the correlation between to the "Variables" field.

2) Tick **Pearson** under "Correlation Coefficients" when the data are continuous and the relationship looks linear, choose Spearman for non-linear relationships or ordinal data.

3) The default for "Tests of Significance" is **Two-tailed**. You could change it to **One-tailed** if you have a directional hypothesis.

4) Under "Options" you can request e.g. Means and standard deviations to be added to the output.

5) Click "OK" to produce the result

19.2 Linear regression

To estimate a linear regression model choose Analyze>> Regression>> Linear from the Menu bar. The following dialog window will appear.

The screenshot shows the 'Linear Regression' dialog box in SPSS. The 'Dependent' field contains 'I will like statistics [Q3]' (annotated with 1). The 'Independent(s)' field contains three variables: 'I am interested in understand...', 'I will understand statistics e...', and 'How good in mathematics ar...' (annotated with 2). The 'Method' is set to 'Enter'. On the right, the 'Statistics...' button is highlighted (annotated with 3), and the 'Plots...' button is highlighted (annotated with 4). The 'Save...' button is highlighted (annotated with 5). At the bottom, the 'OK' button is highlighted (annotated with 6). The list of variables on the left includes 'Number [NR]', 'Kod [Kod]', and various survey items.

1) Add your dependent/response variable to the "Dependent" field.

2) Add your explanatory/independent variable(s) to the "Independent(s)" field.

3) Click "Statistics", tick "Estimates" and "Confidence Intervals" under "Regression coefficients" in the dialog window that will appear. Also ensure that "Model fit" is marked.

4) Click "Plots" and select "Normal Probability plot" under "Standardized Residual Plots".

5) To request residuals used to check the assumptions of linear regression, click "Save", tick "Unstandardized" predicted values and "Studentized" residuals.

6) Click "OK" to produce the result

Parts of the output are explained below.

- 1) These are the values for the regression equation, i.e. the estimated regression coefficients that can be used to interpret the effect the explanatory/independent variable has on the response/dependent variable.

Coefficients ^a								
Model		Standardized Coefficients		Standardized Coefficients	t	Sig.	95,0% Confidence Interval for B	
		B	Std. Error	Beta			Lower Bound	Upper Bound
1	(Constant)	,724	,711		1,018	,310	-,679	2,127
	I am interested in understanding statistical information	,239	,097	,169	2,479	,014	,049	,430
	I will understand statistics equations	,230	,097	,171	2,382	,018	,040	,421
	How good in mathematics are you?	,352	,104	,235	3,366	,001	,146	,557

a. Dependent Variable: I will like statistics

2) Beta –standardized regression coefficients. They can be used to assess which of the explanatory variables that have the largest effect on the response variable, after taking into account that variables are measured on different scales.

3 & 4) These are the t-statistics and their associated 2-tailed P-values used in testing whether a given coefficient is significantly different from

5 & 6) 95% confidence intervals for the regression coefficients

- 1) **R** –for simple regression this is the correlation between the explanatory and response variable.
 2) **R Square** – the coefficient of determination. This explains how much of the variation in the response variable that can be explained by the different values of the explanatory variable(s).

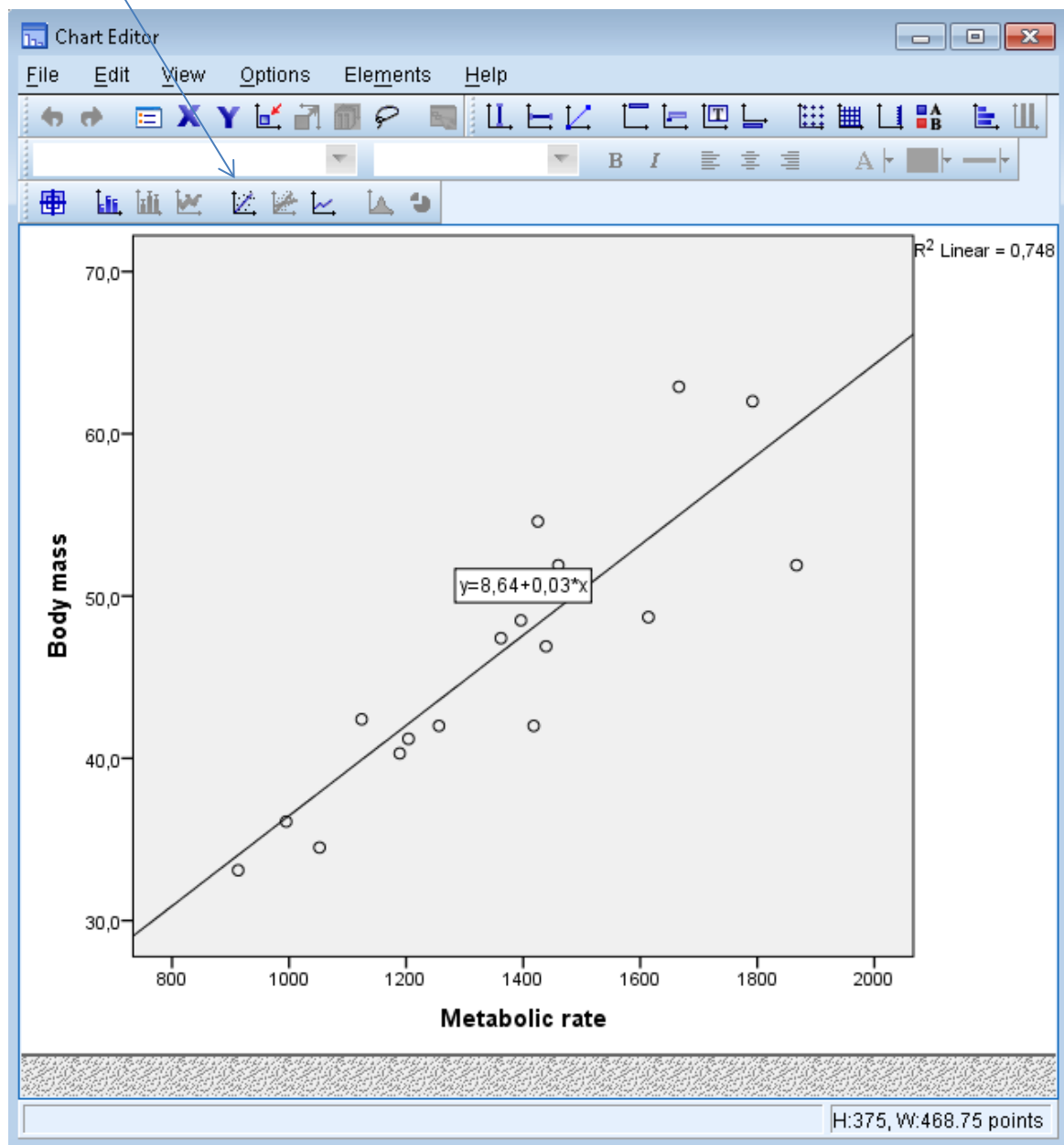
Model Summary ^b				
Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	,410 ^a	,168	,155	1,694

3) Std. Error of the Estimate – standard error of the regression prediction, i.e. the average distance from the regression line.

19.2.1 Add a regression line to scatterplot

To add a regression line to a scatterplot, start by producing a simple scatter plot as described in Section 9.8.1.

In the output window, double click the obtained graph to open the Graph Editor. Click “Add a fit line” and ensure that “Linear” is marked under “Fit method” in the dialog window that will appear. Then click “Close”.



20 Logistic regression

To estimate a Binary Logistic Regression choose **Analyze >> Regression >> Binary Logistic** from the Menu tab. The dialog window below will appear.

The image shows the SPSS Logistic Regression dialog box with four numbered annotations:

- 1)** Add the response variable to the "Dependent" field. (Gender [Gender])
- 2)** Add the explanatory variables to the "Covariates" field. (Q28, Q31)
- 3)** Click "Options" and select "Hosmer-Lemeshow Goodness of fit" and "CI for exp(B)".
- 4)** Click "OK" to produce the result.

The dialog box includes a list of variables on the left, a "Dependent" field, a "Covariates" field, a "Method" dropdown (set to Enter), and buttons for "Previous", "Next", "Categorical...", "Save...", "Options...", "Style...", "OK", "Paste", "Reset", "Cancel", and "Help".

Parts of the output are explained below.

1) B - this is the coefficient for the constant (intercept) in the null model.

This output is for the Block= 0, which describes a "null model", the model with no predictors, just the intercept.

Variables in the Equation											
	1	B	2	S.E.	3	4	Wald	df	5	Sig.	Exp(B)
Step 0 Constant		,031		,143			,046	1		,830	1,031

2) S.E – the standard error around the coefficient for the constant.

3, 4) Wald and Sig -this is the Wald Chi-Square that tests the null hypothesis that the constant equals 0.

5) Exp(B) –the exponentiation of the B coefficient, which is an odds ratio.

Variables in the Equation							
	1	2		3	4	5	
	B	S.E.	Wald	df	Sig.	Exp(B)	
Step 1 ^a Q28	-,246	,075	10,68	1	,001	,782	This is usually the interesting part of the output
Q31	-,012	,144	,007	1	,933	,988	
Constant	1,013	1,019	,989	1	,320	2,755	

1) B –the values for the logistic regression equation for predicting the dependent variable from the independent variable in terms of the original coefficients, i.e. **log(odds)**.

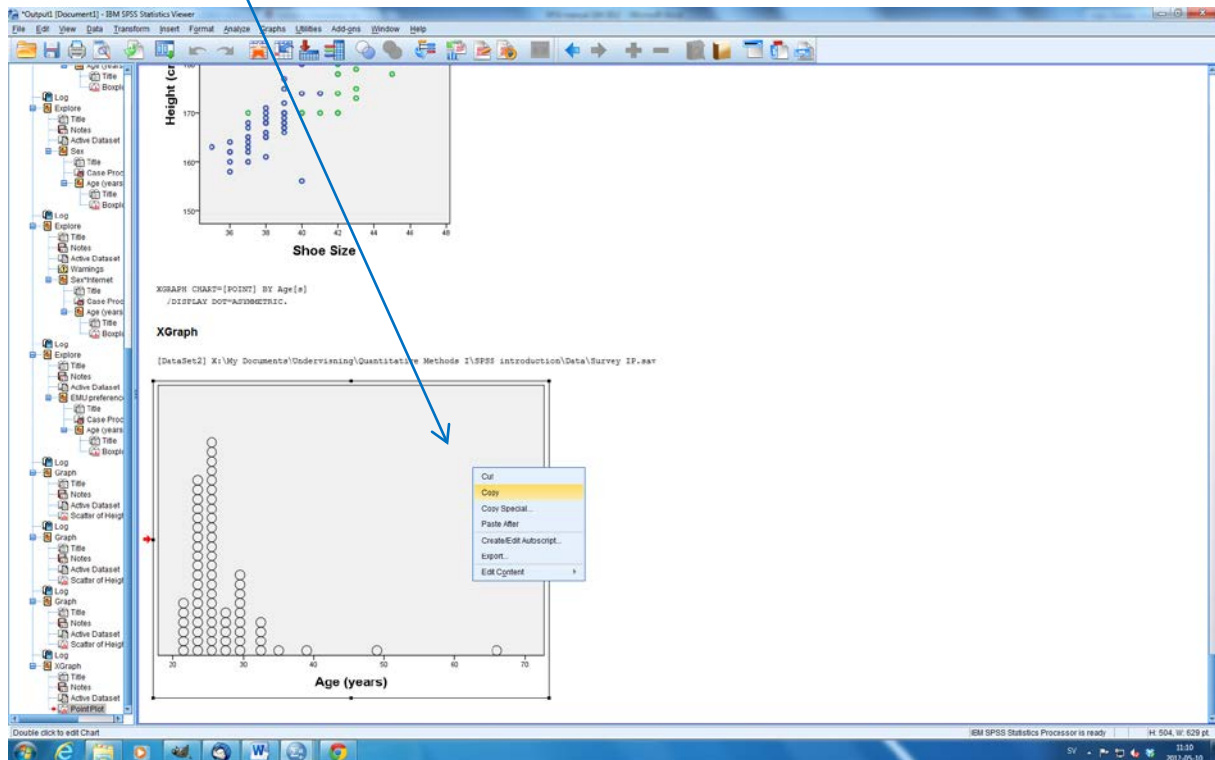
2) S.E – the standard errors associated with the coefficients.

5) Exp(B) – **Odds ratios**. Tells how many times the odds of the event of interest changes when the explanatory variable increases by 1 unit.

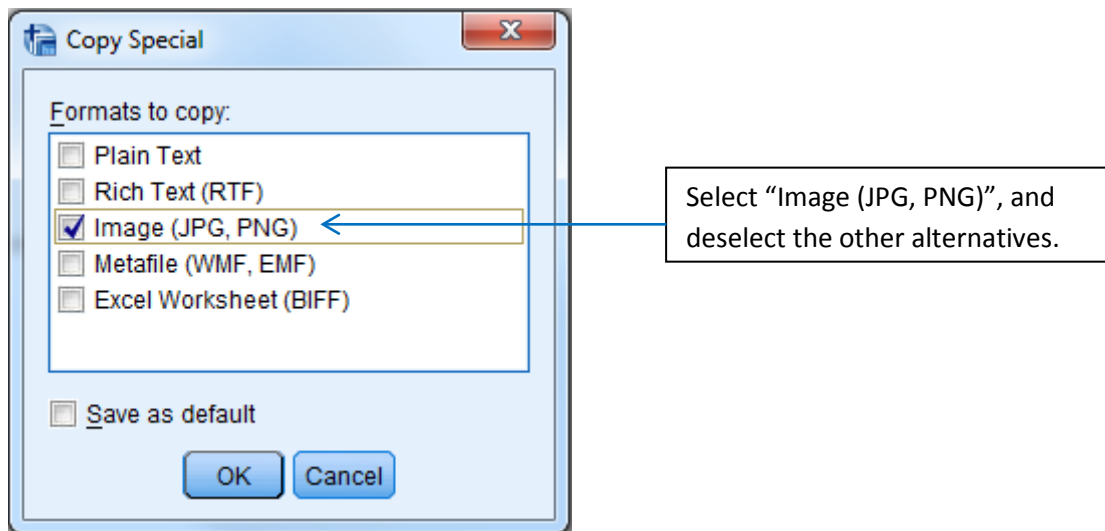
3, 4) Wald and Sig - These columns provide the Wald chi-square value and 2-tailed **P-value** used in testing the null hypothesis that the coefficient (parameter) is 0, or equivalently that the Odds Ratio is 1.

21 Copying output to Word

To copy output to e.g. Word, right click the output (graph, table, etc.) in the Output window and choose **Copy**. Then paste it into a Word document (e.g. by using the keyboard shortcut Ctrl+V)



If this doesn't work, you can try choosing **Copy Special** instead. The dialog window below will then appear.



Click "OK", and then paste it into a Word document (e.g. by using the keyboard shortcut Ctrl+V).

22 Copying output to PowerPoint

To copy output to PowerPoint, follow the instructions in section 21 above. In some versions of PowerPoint you always have to choose **Copy Special**.